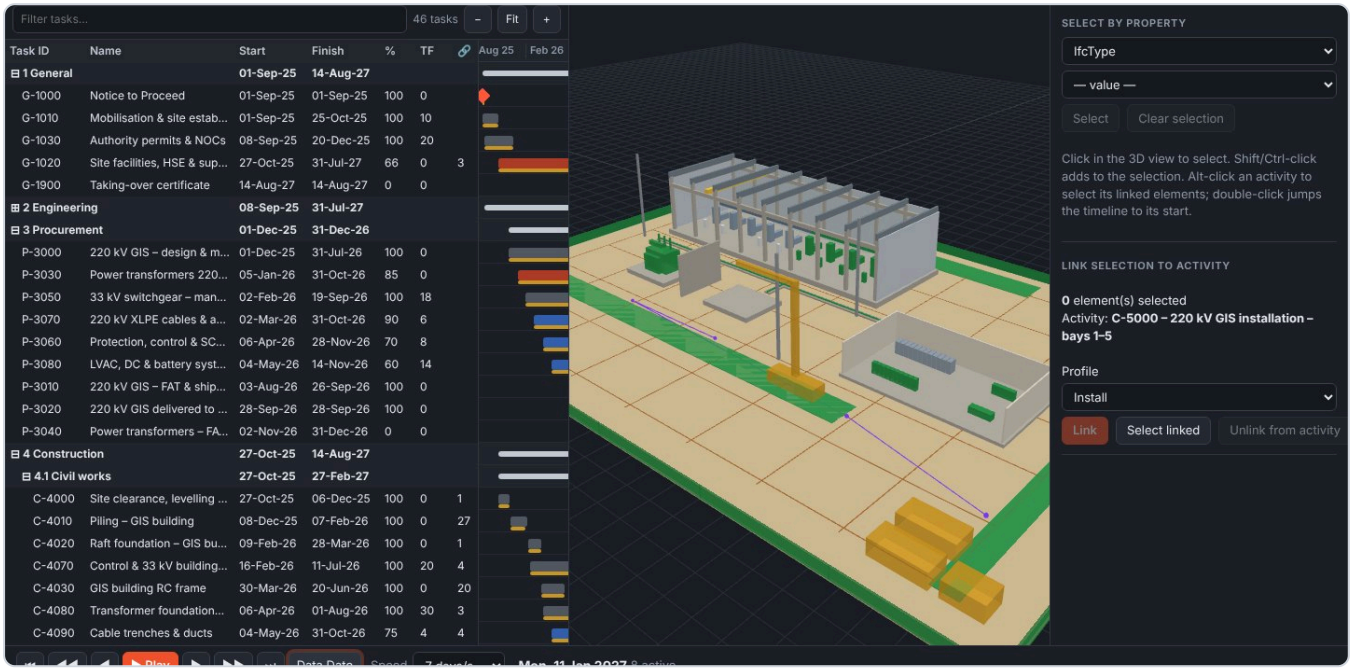
User Guide

Release 0.3 · Interface layouts, 4D simulation, project database, Primavera P6 connections, dashboards and emailed reports



Worked example: 220/33 kV GIS Substation, Abu Dhabi (fictional sample project)

Conceived, designed and developed by

Junaid Quamar

Senior Planning Engineer · linkedin.com/in/junaidquamar

User guide by Junaid Quamar
October 2026

Contents

About the author

1. About this guide

2. Installing Stratum

3. First start

4. The three layouts

5. Working with the schedule

6. 4D simulation

7. Cost, resources and quantities

8. The project database

9. Primavera P6 connections

10. Dashboards and reports

11. Stratum Server for teams

12. Keyboard shortcuts

13. Troubleshooting

14. Known limits

Appendix A · Example project activities

Appendix B · Terms in each layout

About the author

Junaid Quamar

Senior Planning Engineer

Stratum 4D was conceived, specified, designed and directed by Junaid Quamar. The feature set, its priorities and the planning workflows it supports come from his project-controls practice in Primavera P6 planning and cost engineering for HV substation, transmission and rail electrification projects.

That background shows throughout the tool: the P6 and MS Project layouts planners already know, a CPM engine that follows P6's scheduling rules, two-way P6 connections, EPC-style weighted progress reporting with manual weightage, and the 220 kV GIS substation used as the worked example in this guide.

This guide was written and prepared by Junaid Quamar.

Name	**Junaid Quamar**
Role	Senior Planning Engineer
Focus	Primavera P6 planning, project controls and cost engineering; HV substations, transmission and rail electrification
LinkedIn	linkedin.com/in/junaidquamar
Product	Stratum 4D · release 0.3 · October 2026

> Credit: Stratum 4D and this user guide were developed by Junaid Quamar. Please keep this credit when sharing or adapting the guide.

1. About this guide

This guide covers Stratum 4D release 0.3: the three interface layouts (P6, MS Project and 4D), scheduling, 4D simulation, the project database, Primavera P6 connections, and the dashboard and report module with emailed reports. Every screenshot comes from the application itself, running as a Stratum Server, with one example project loaded throughout.

The example project

The example is a **220/33 kV GIS substation in Abu Dhabi**, delivered as an EPC contract. It ships with Stratum as a sample (*File → Sample: 220 kV GIS Substation (UAE)*, or the button on an empty screen), so you can follow every step yourself. The project is fictional. Names, quantities and values are illustrative, and the Islamic public holidays in its calendar are approximate because they follow the moon.

Item	Example project
Project	GIS-220 · 220/33 kV GIS Substation, Abu Dhabi (sample project)
Scope	220 kV GIS hall with 9 bays and 10 t EOT crane; two 220/33 kV power transformers with firewall and oil pits; control & 33 kV building (33 kV switchgear, protection, control & SCADA, LVAC, DC); cable trenches and 220 kV XLPE cables; earthing grid and lightning masts; roads, drainage, boundary wall and gatehouse
Contract structure (WBS level 1)	General · Engineering · Procurement · Construction (Civil works, Electromechanical works, Testing & commissioning)
Schedule	46 activities, 62 relationships (FS, SS and FF with lags), activity codes *Discipline* and *Area*, user field *Package*
Calendar	UAE 6-day week, Monday–Saturday, 07:00–16:00 with a one-hour break; UAE public holidays 2025–2027
Resources and cost	15 resources in AED: labour (design engineers, civil crew, electrical technicians, OEM specialists, testing engineers), plant (200 t mobile crane, piling rig) and procured equipment (GIS, transformers, switchgear, panels, cables)
Data date	19-Oct-26
3D model	202 elements carrying the property `Pset_Stratum.ActivityID`, so one rule links the whole model

Status at the data date

Engineering is complete apart from the as-built documentation. The 220 kV GIS arrived on site on 28-Sep-26, three weeks after its baseline date, and installation of bays 1–5 has started. The power transformers are 85% through manufacture and running 21 days late, which puts transformer erection, pre-commissioning and energisation on the critical path. The forecast taking-over date is 14-Aug-27 against a baseline of 31-Jul-27.

Milestone	Name	Current / forecast	Baseline	Status
G-1000	Notice to Proceed	01-Sep-25	01-Sep-25	Achieved
P-3020	220 kV GIS delivered to site	28-Sep-26	07-Sep-26	Achieved

Milestone	Name	Current / forecast	Baseline	Status
C-6010	220 kV energisation	31-May-27	17-May-27	Forecast

> Critical path after rescheduling (F9): `G-1020` → `G-1900` → `P-3030` → `P-3040` → `C-4900` → `C-5030` → `C-5040` → `C-6000` → `C-6010` → `C-6020`. CPM finish: 12-Aug-27.

2. Installing Stratum

Stratum runs three ways. The engine and the screens are the same in all of them; what changes is where projects are stored and which services are available.

Way to run	Where projects live	P6 connections	Emailed reports	Typical use
Desktop app (Windows, macOS, Linux)	A local SQLite database in your user folder	Yes	Yes, while the app is open	One planner, offline work
Browser	The browser's own storage	No (use XER files)	No	Quick look, demos
Stratum Server (shared)	One PostgreSQL database for the team	Yes	Yes, around the clock	Project controls team

Desktop installers

Installers are built for every release: an NSIS installer for Windows, DMG files for Intel and Apple Silicon Macs, and AppImage/deb packages for Linux. They are produced by the project's GitHub Actions workflow and appear under *Actions → Desktop installers → latest run → Artifacts*.

The installers are not code-signed yet. On Windows, SmartScreen shows "Windows protected your PC": choose *More info → Run anyway*. On a Mac, right-click the app and choose *Open* the first time. The Windows installer lets you choose the install folder and adds Start-menu and desktop shortcuts. Double-clicking an `.xer` or `.ifc` file opens it in Stratum.

Stratum Server

See Stratum Server for teams for the Docker Compose setup, environment settings, users and roles.

3. First start

The first time Stratum opens, a short setup asks two questions. Both can be changed later in Settings.

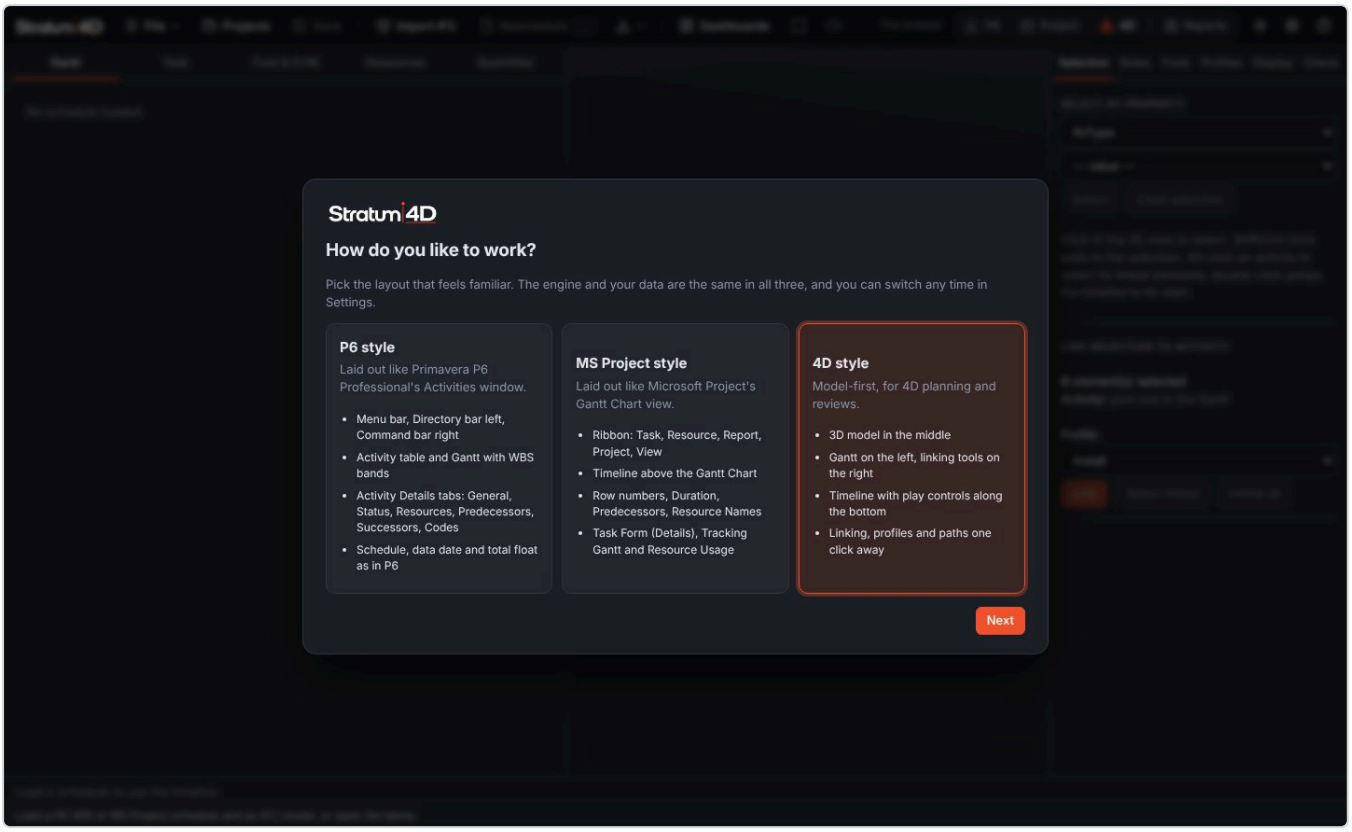


Figure 1. Step 1: choose the layout that feels familiar. The data and every feature are the same in all three.

Layout. *P6 style* is laid out like P6 Professional's Activities window, *MS Project style* like Microsoft Project's Gantt Chart view, and *4D style* puts the 3D model in the middle for 4D planning and reviews. Section 4 describes each layout.

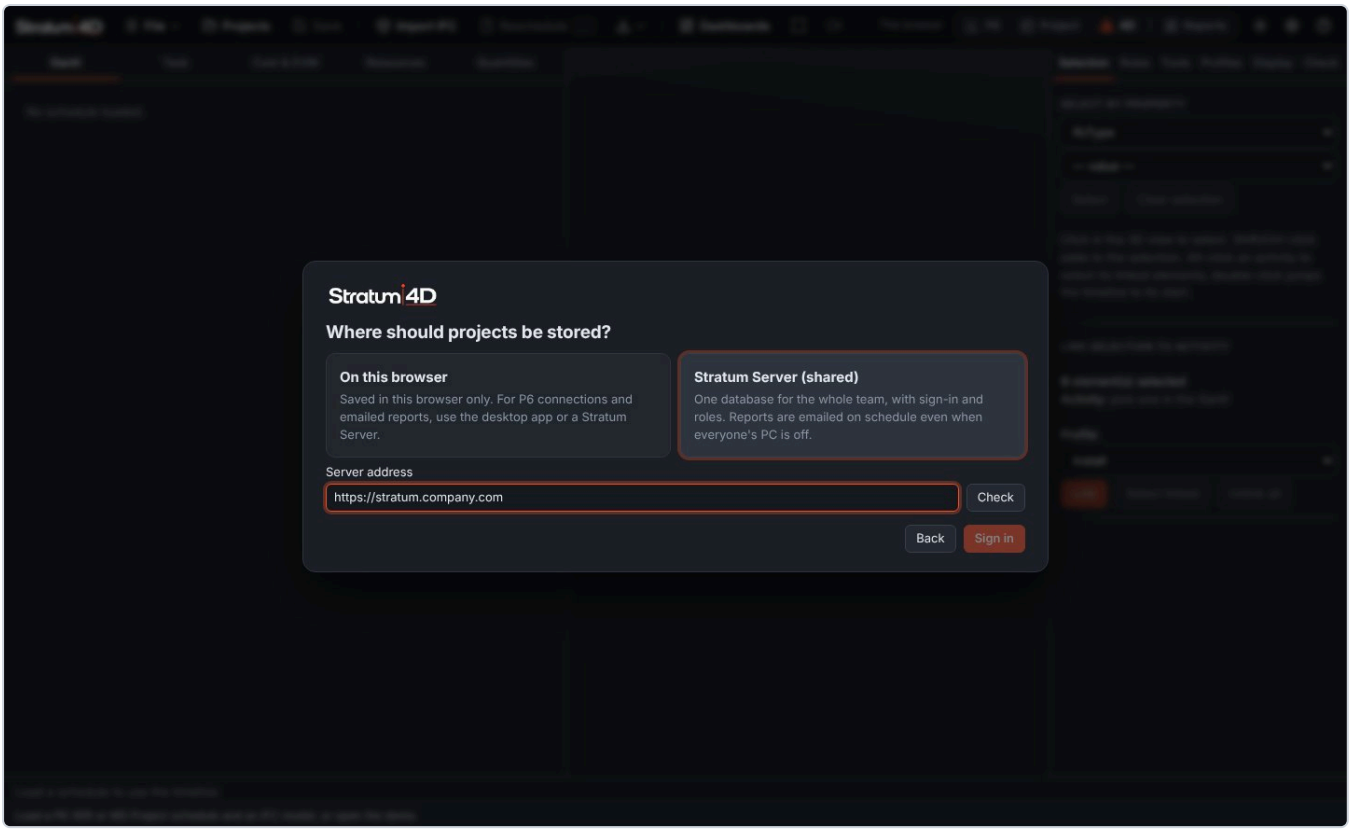


Figure 2. Step 2: where projects are stored. Choosing Stratum Server asks for the server address, then for an account; on a brand-new server the first account becomes the administrator.

Storage. In the browser the choice is between this browser and a Stratum Server. In the desktop app the first option is the local database on your computer. After setup Stratum offers to open the P6 and email settings, since both are usually configured once by an administrator.

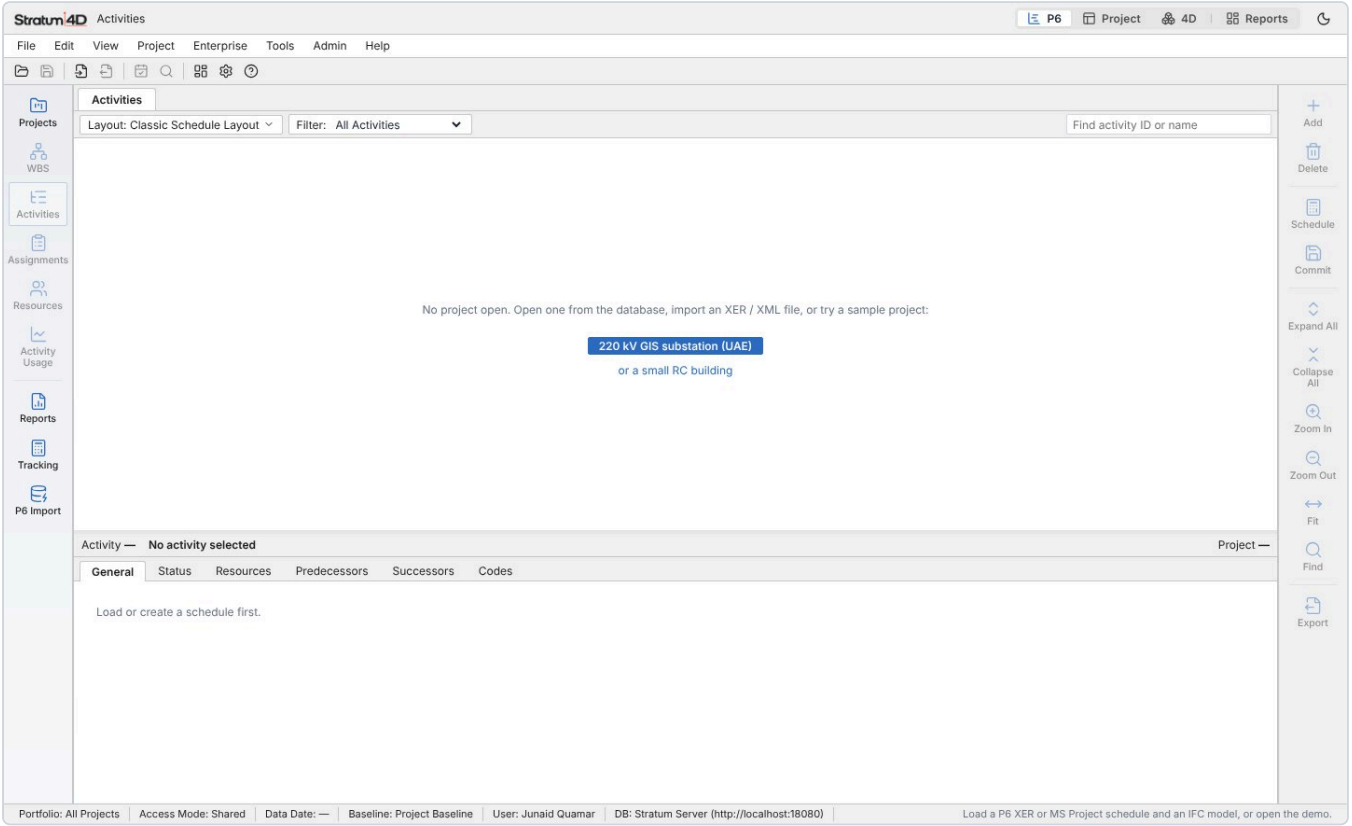


Figure 3. An empty P6 layout. The buttons in the middle load one of the two built-in sample projects.

4. The three layouts

Stratum offers three layouts over the same engine and data. The **P6** and **MS Project** layouts are schedule-only, laid out like the applications planners already know, with no 3D view. The **4D** layout adds the model, linking tools and playback. The open project, the selection and the filters stay as they are when you switch.

Switching layouts

Click **P6**, **Project** or **4D** at the top right, or press `Ctrl` + `Alt` + `1`, `2` or `3`. The fourth button, **Reports** (`Ctrl` + `Alt` + `4`), opens the dashboard and report builder; the same switch sits in its top bar, so one click takes you back to any layout. The sun/moon button beside it switches between light and dark.

	P6 style	MS Project style	4D style
Laid out like	Primavera P6 Professional, Activities window	Microsoft Project, Gantt Chart view	A 4D planning tool
Top of the screen	Title bar, menu bar (File, Edit, View, Project, Enterprise, Tools, Admin, Help), icon toolbar	Green title bar with quick-access buttons, ribbon tabs (File, Task, Resource, Report, Project, View) and groups	Toolbar with icons and labels
Sides	Directory bar left, Command bar right	Vertical view label (Timeline, Gantt Chart, Task Form)	Gantt left, linking tools right
Main area	Layout and filter bar; activity table and Gantt with WBS bands	Timeline; Gantt Chart with row numbers, indicators, task mode, predecessors and resource names	3D model with the Gantt and timeline
Lower pane	Activity Details tabs, or a Resource / Activity Usage Profile	Task Form (Details), optional	Timeline playback
Gantt bars	Actual blue, remaining green, critical red; black WBS summary bars	Blue task bars with a progress line; black summary brackets; milestone dates; Tracking Gantt shows critical work and baselines	Blue bars, red critical, playback line
3D view	No	No	Yes

P6 style

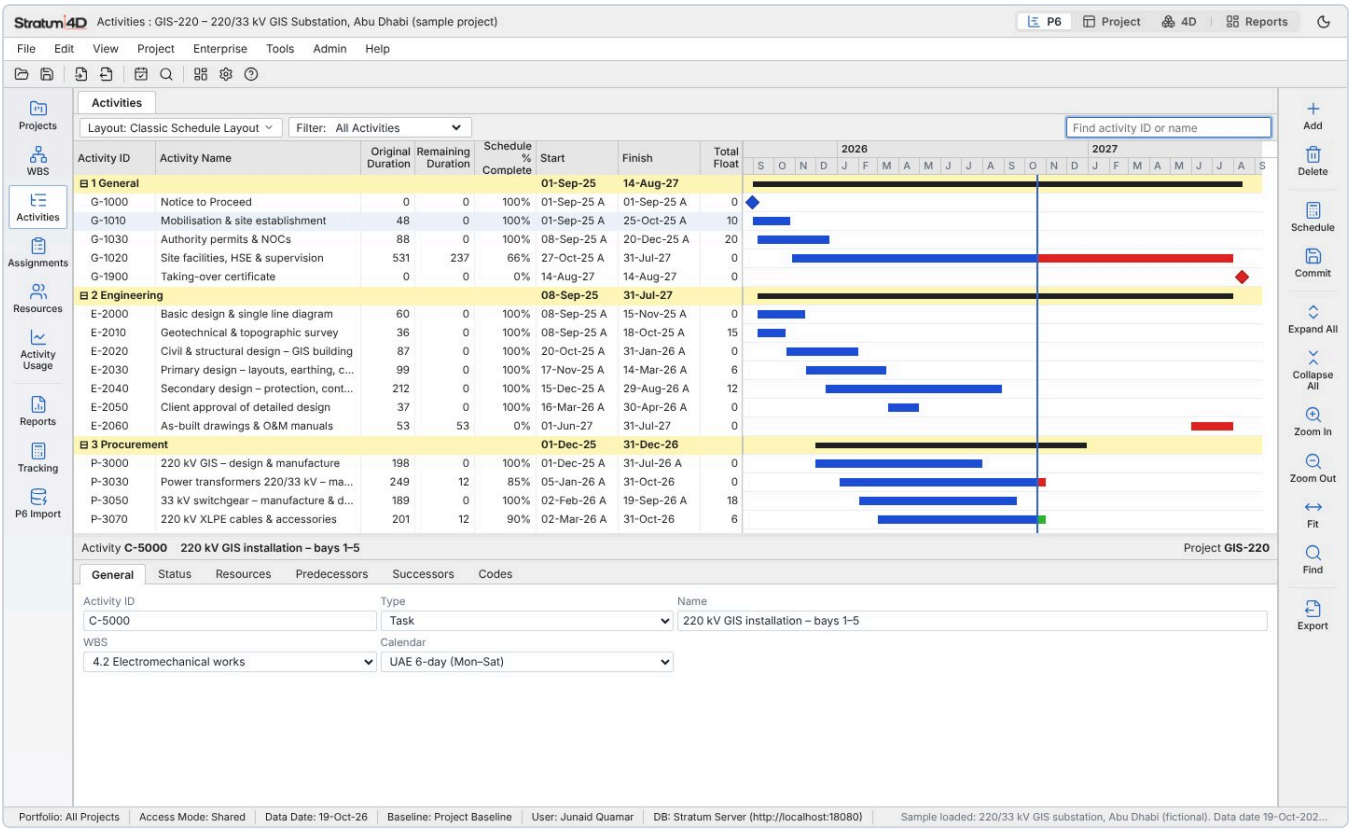


Figure 4. The P6 layout with the example project. C-5000 (220 kV GIS installation, bays 1–5) is selected and its General tab is open. WBS bands are coloured by level, actual dates carry an "A", and the vertical blue line is the data date, 19-Oct-26.

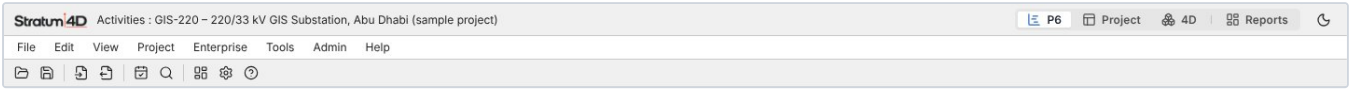


Figure 5. Title bar, menu bar and toolbar: the layout switch with Reports at the top right, then open, commit, import, export, schedule (F9), find, reports, preferences and help.

The **Directory bar** on the left decides what you look at: Projects, WBS, Activities, Assignments, Resources (Resource Usage Profile), Activity Usage (cost and earned value), Reports, Tracking and P6 Import. The **Command bar** on the right holds what you do: Add, Delete, Schedule, Commit, Expand All, Collapse All, Zoom In, Zoom Out, Fit, Find and Export.

Above the table, *Layout* switches between the Classic Schedule Layout and the same layout with baseline bars, and *Filter* shows all activities or only critical, not started, in progress, completed, late or milestone activities. The find box filters by activity ID or name. The table has P6's Classic Schedule Layout columns: Activity ID, Activity Name, Original Duration, Remaining Duration, Schedule % Complete, Start, Finish and Total Float.

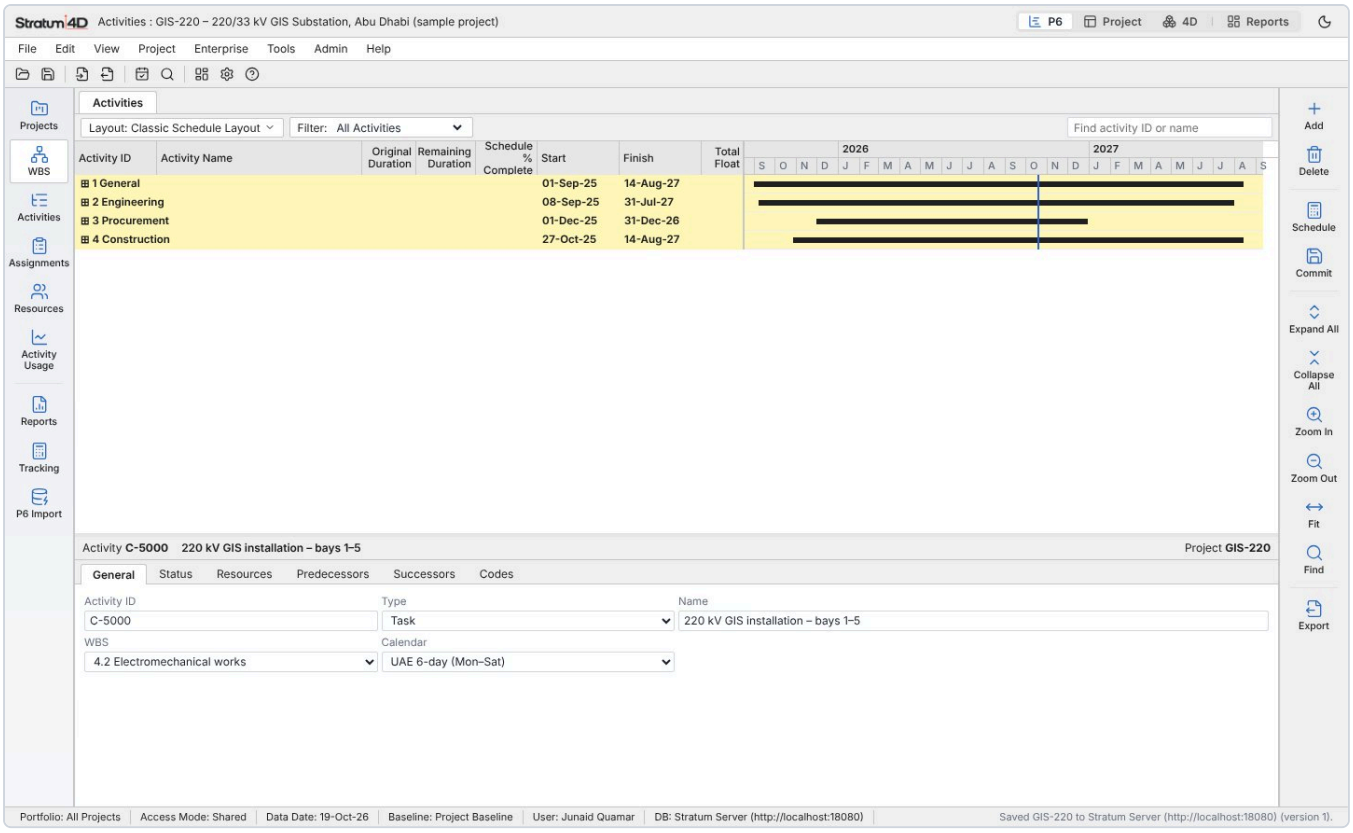


Figure 6. WBS on the Directory bar collapses the table to WBS bands: General, Engineering, Procurement and Construction with their dates. Activities expands it again.

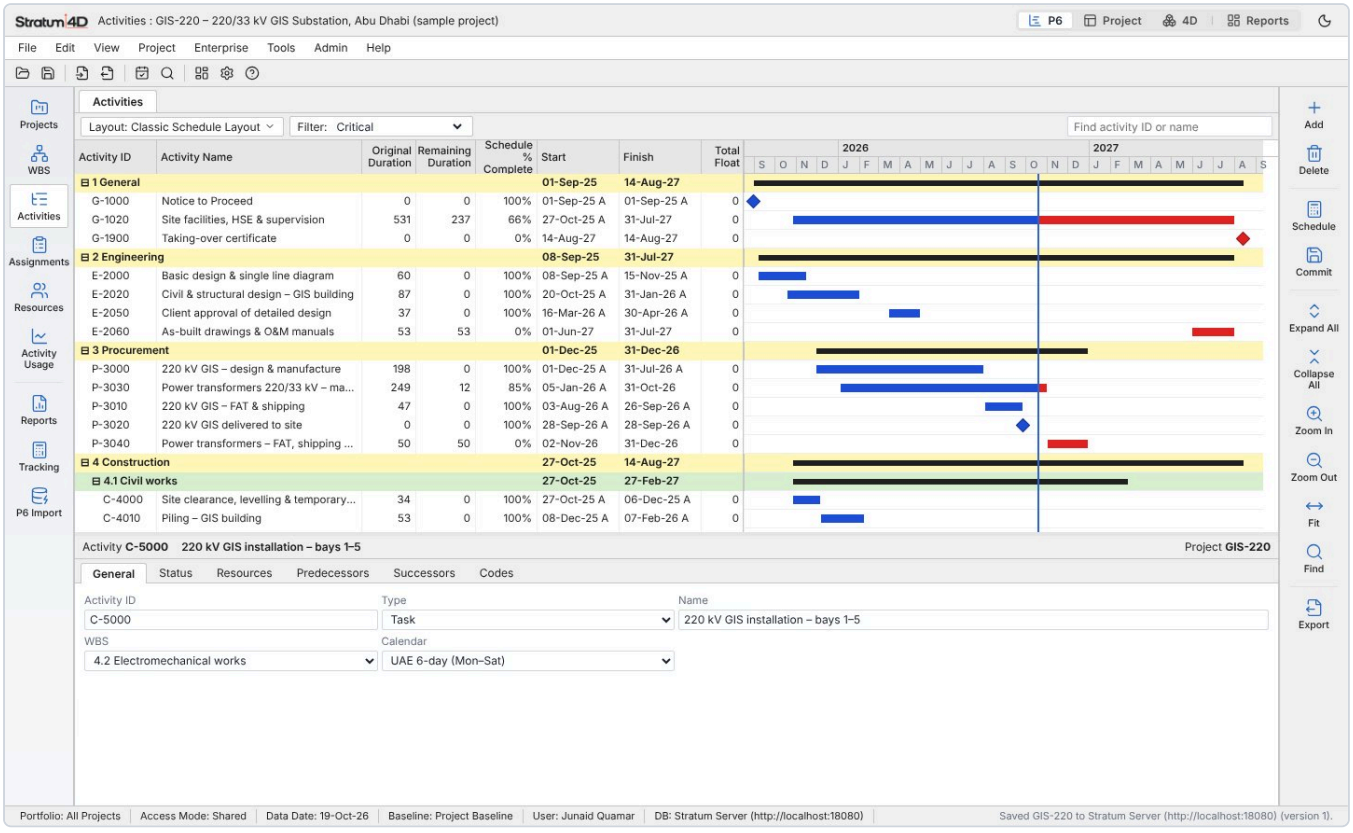


Figure 7. Filter: Critical keeps the critical activities with their WBS bands: transformer manufacture and delivery, transformer placing and erection, testing and energisation.

MS Project style

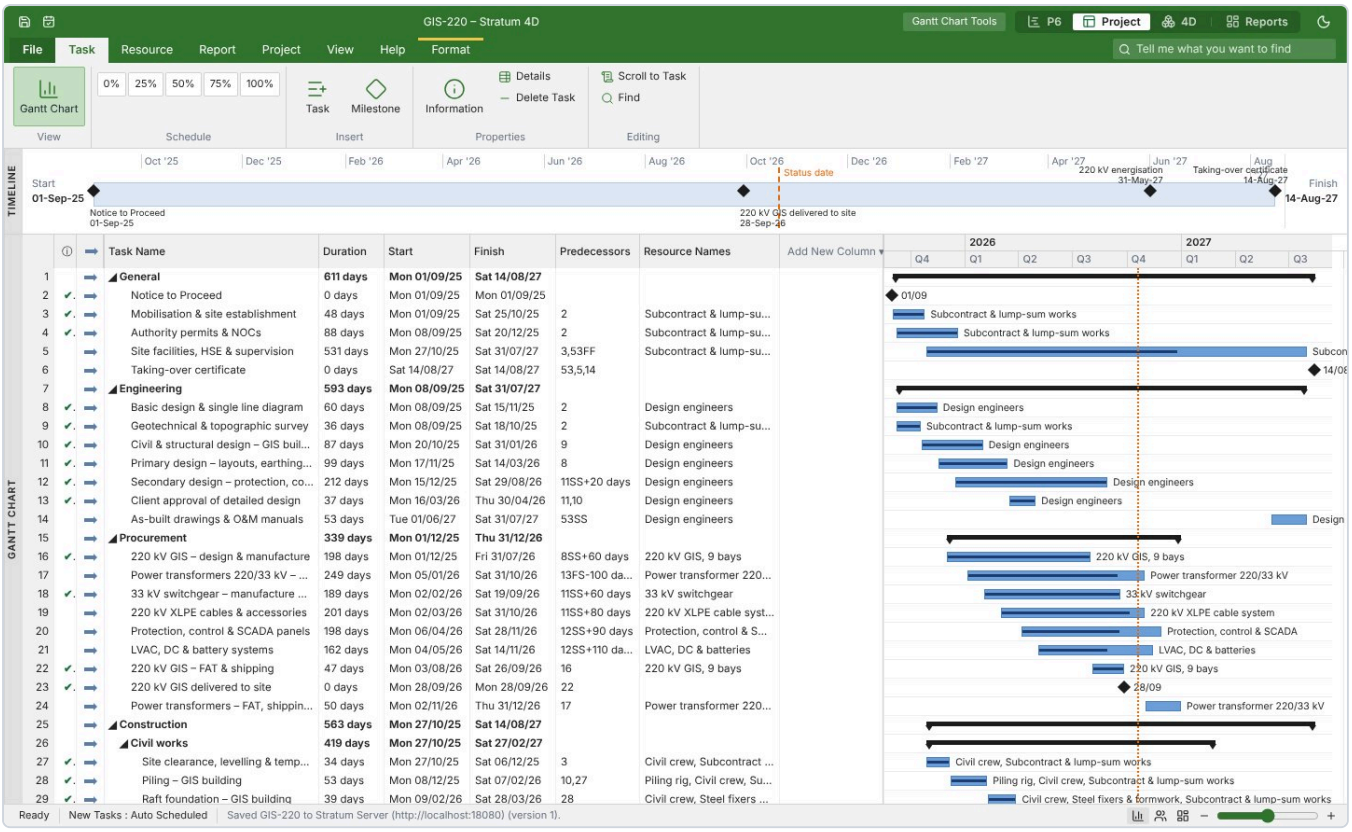


Figure 8. The MS Project layout: ribbon with the Task tab open (Gantt Chart, % complete buttons, insert Task or Milestone, Information, Details, Scroll to Task, Find), the Timeline with the status date and milestones, and the Gantt Chart with resource names beside the bars.

The Task tab's *0% – 100%* buttons set the selected task's progress, and *Calculate Project* on the Project tab reschedules (F9). *Details* opens the Task Form under the Gantt; *Tracking Gantt* on the View tab shows critical work in red with the baseline under each bar.

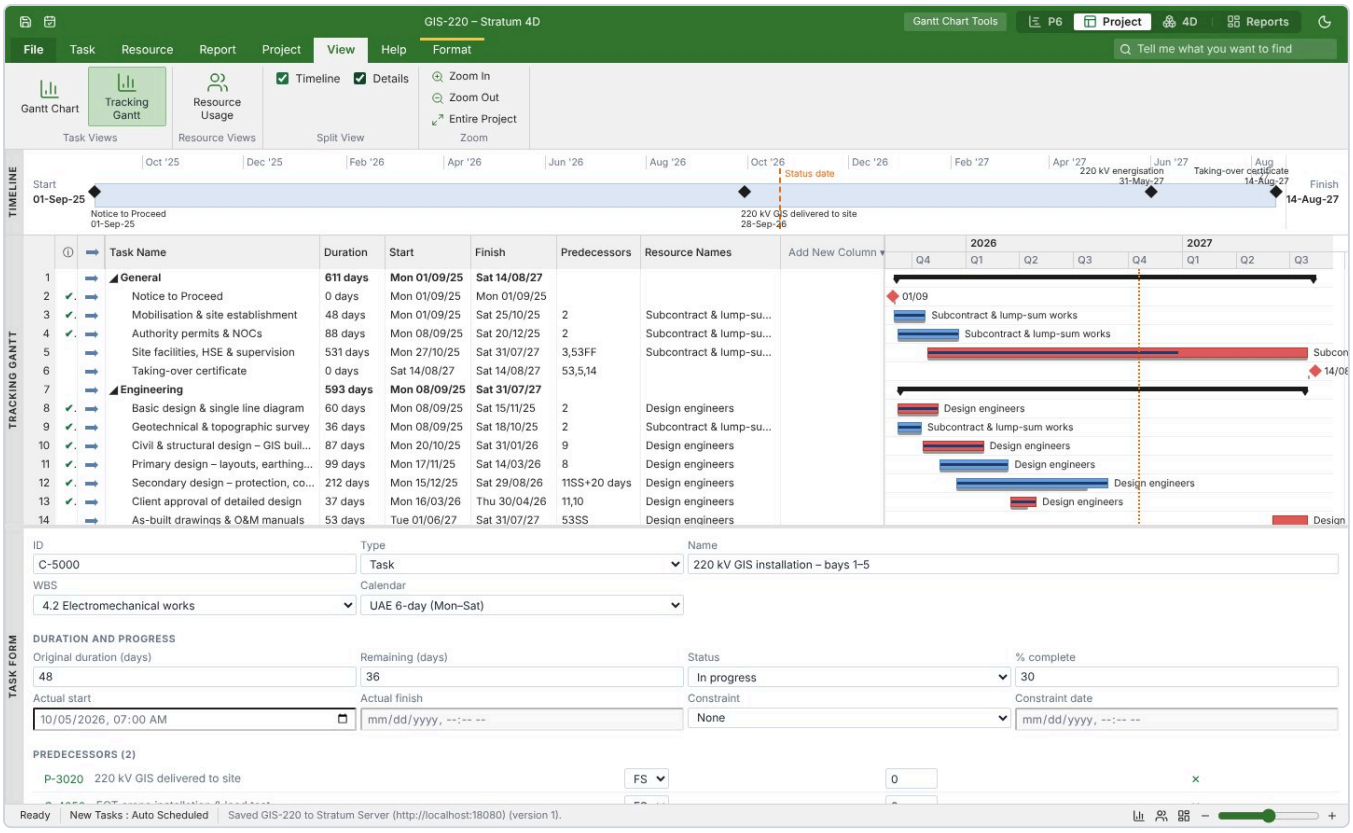


Figure 9. View → Tracking Gantt with Details ticked: critical tasks in red, baselines under the bars, and the Task Form below.

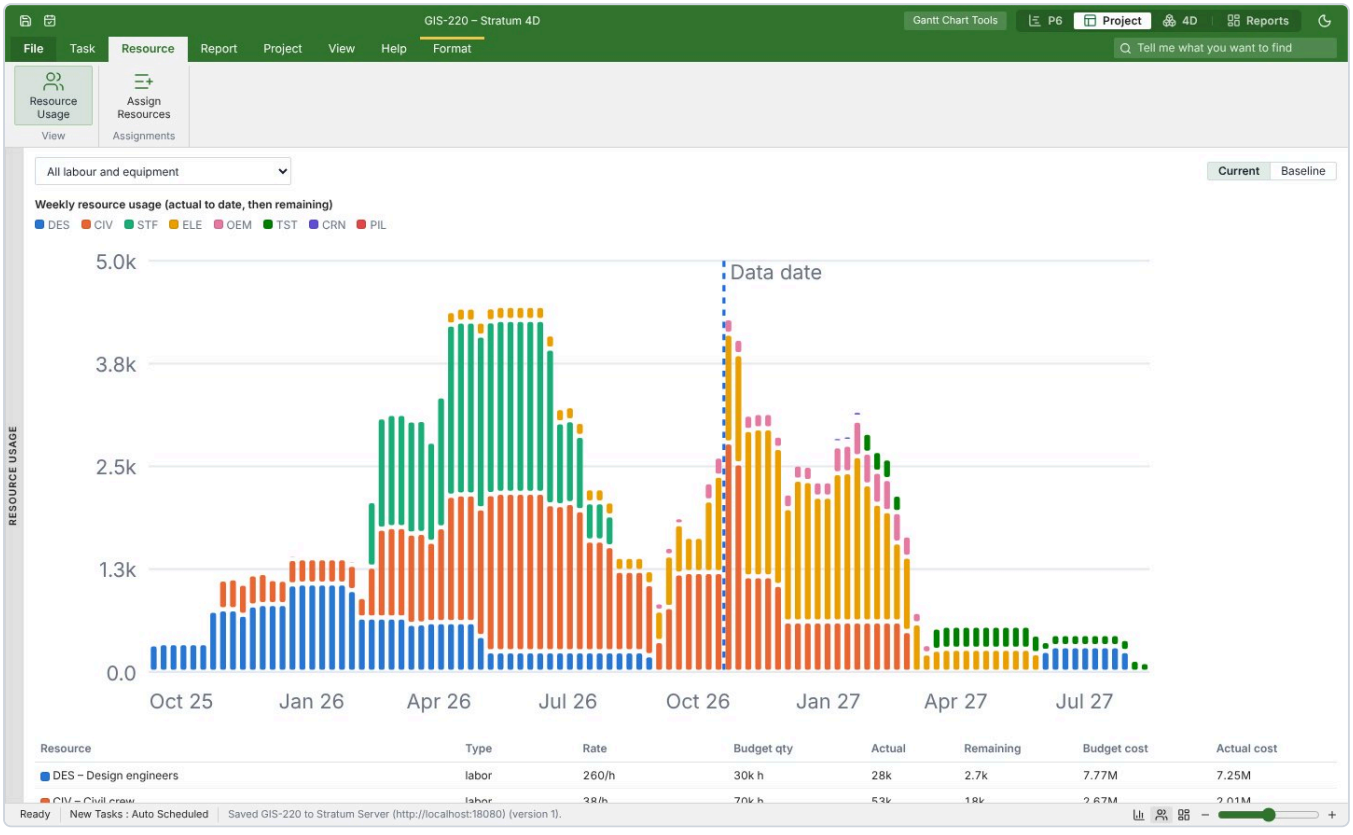


Figure 10. Resource → Resource Usage: weekly hours per resource, actual to date then remaining, with a table of budget, actual and remaining quantities and costs.

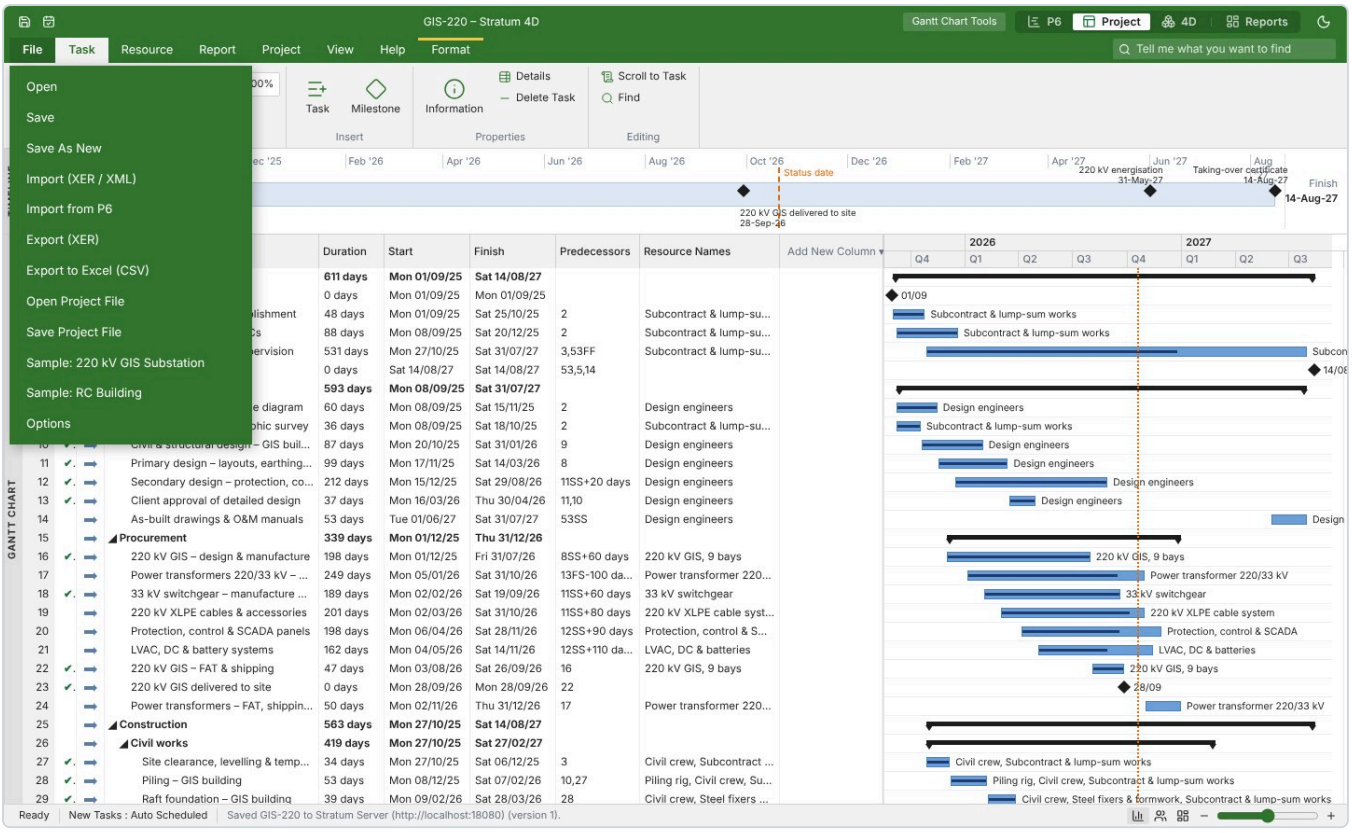


Figure 11. The File tab: open and save in the database, import and export, project files, the samples and options.

4D style

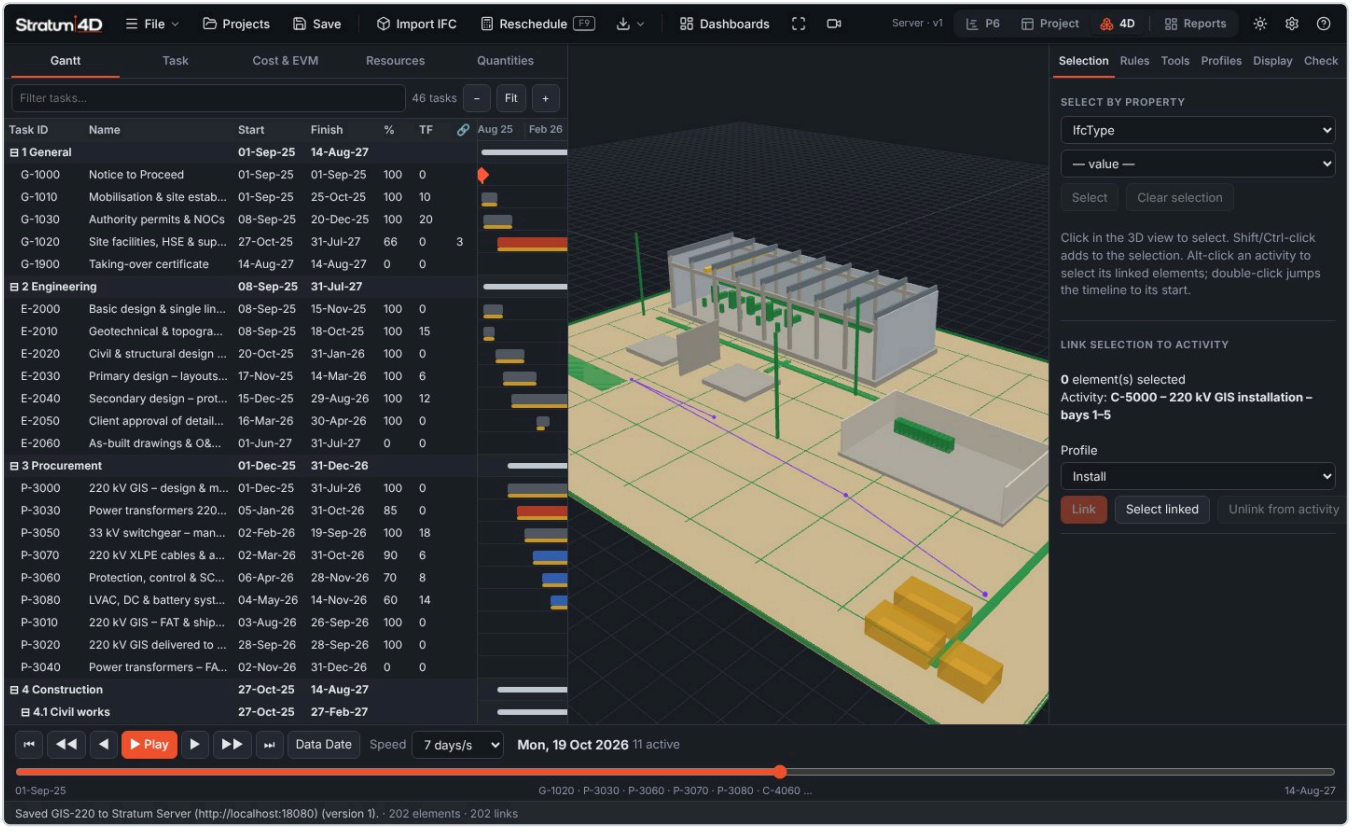


Figure 12. The 4D layout in its default dark theme. Green elements are being installed at the data date: GIS bays, cable trenches and 33 kV switchgear. The amber boxes are the site cabins, shown only while site facilities (G-1020) run.

Section 6 covers the 4D layout in detail.

Theme, font and text size

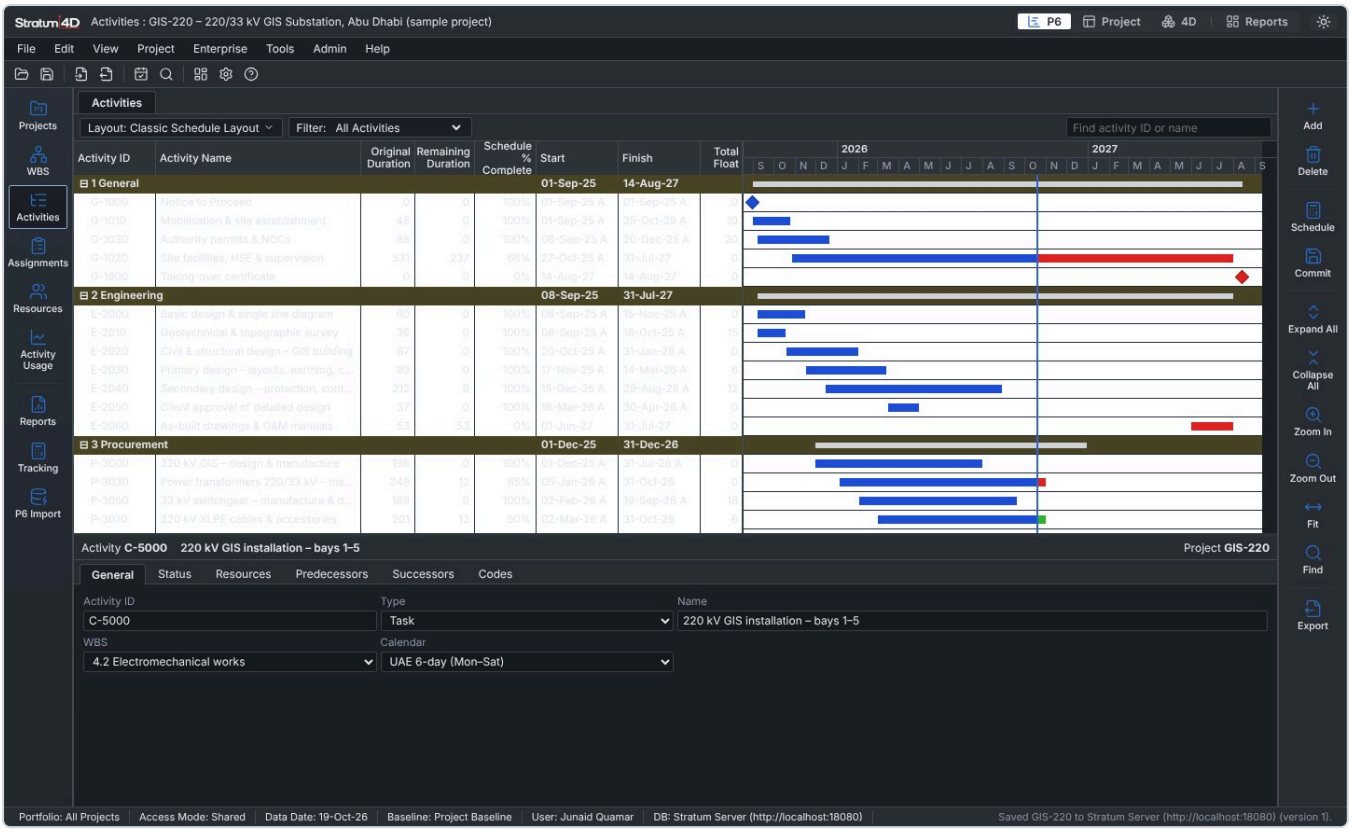


Figure 13. The P6 layout in the dark theme.

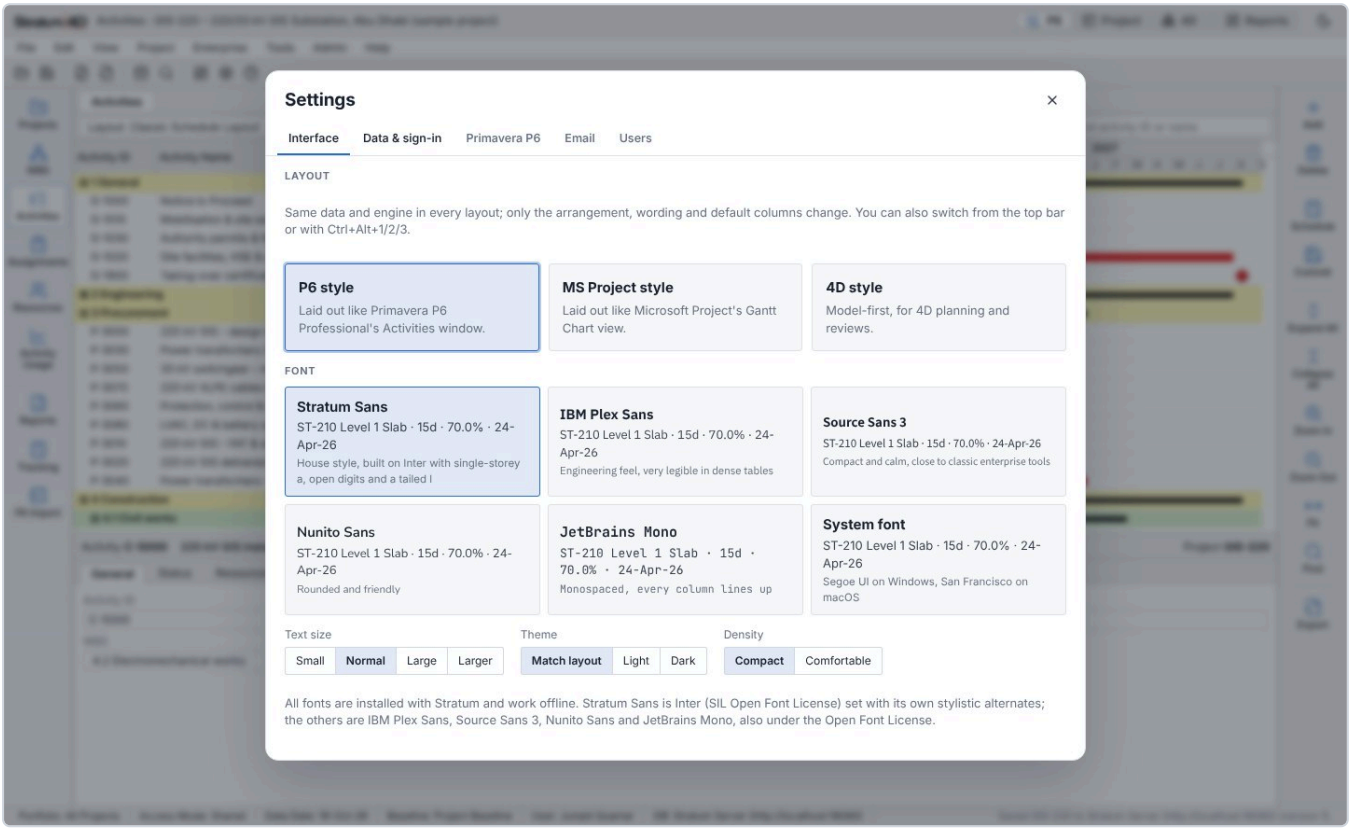


Figure 14. Settings → Interface (in the P6 layout: Admin → Preferences). Each font shows a sample line in its own face.

Fonts are installed with Stratum, so the desktop app looks the same offline. The default, **Stratum Sans**, is Inter set with its own stylistic alternates. IBM Plex Sans, Source Sans 3, Nunito Sans, JetBrains Mono

and the system font are also offered, all under the SIL Open Font License. Table columns widen automatically for larger text or wider fonts.

5. Working with the schedule

Importing

Import reads Primavera P6 XER files and Microsoft Project XML (MSPDI). From XER, Stratum reads WBS, activities, relationships with lags, calendars, constraints, activity codes, user-defined fields, actual and baseline dates, resources, rates and assignments. Re-importing a newer XER over the open project updates it; links to the model are kept by Activity ID.

Activity details

In the P6 layout the lower pane holds the Activity Details tabs for the selected activity: General (ID, type, name, WBS, calendar), Status (durations, status, % complete, actual dates, constraint, early/late dates and float), Resources (assignments with budget, actual and remaining units and cost), Predecessors, Successors and Codes (user fields).

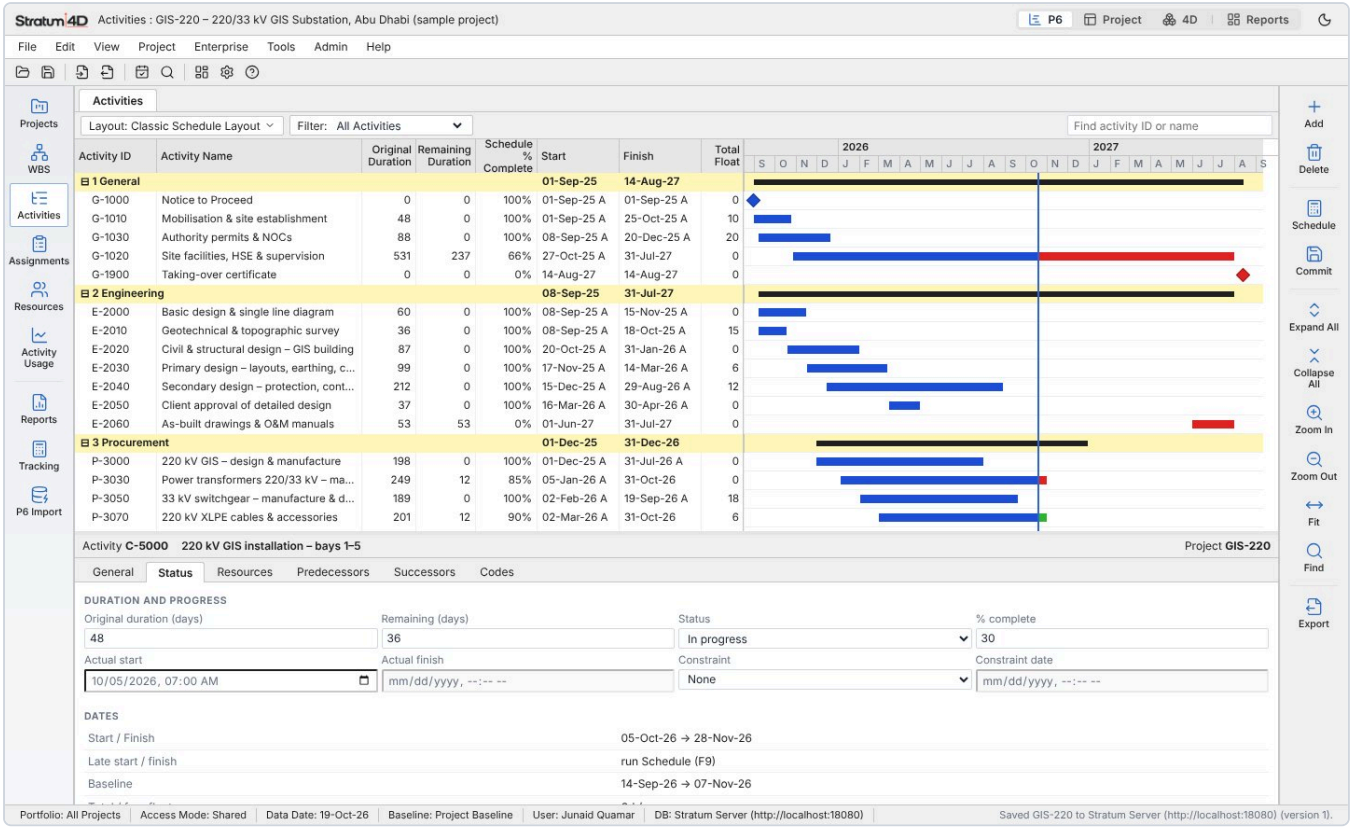


Figure 15. Status tab for C-5000: original 48 days, remaining 36, in progress at 30% with its actual start; the dates section below shows early and late dates and float after scheduling.

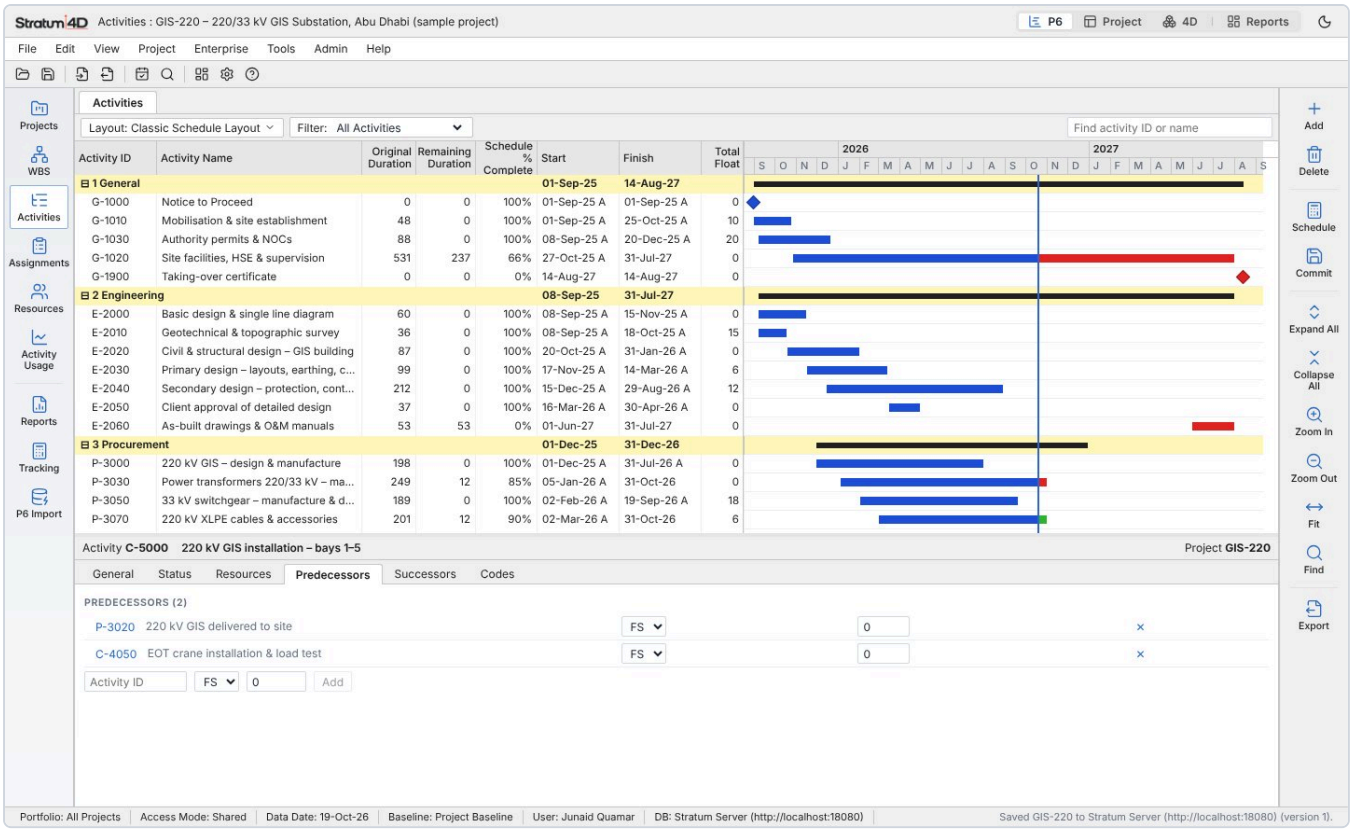


Figure 16. Predecessors tab: relationship type and lag for each predecessor, and a row to add one by activity ID.

Add (`Ins`) inserts an activity after the selected one with a 5-day duration and a finish-to-start link; *Delete* (`Del`) removes it with its relationships. Edits mark the schedule as changed and a banner asks you to reschedule.

Scheduling (F9)

Press `F9`, *Schedule* (P6) or *Calculate Project* (MS Project). The CPM engine follows P6's default options:

- FS, SS, FF and SF relationships with lags, on each activity's working calendar and holidays;
- nine constraint types;
- retained logic from the data date;
- total and free float, critical path, level-of-effort and WBS summary dates;
- loop detection and open-end warnings.

For the example project F9 finishes on 12-Aug-27, with the critical path running through transformer manufacture, delivery, erection and energisation.

Exporting

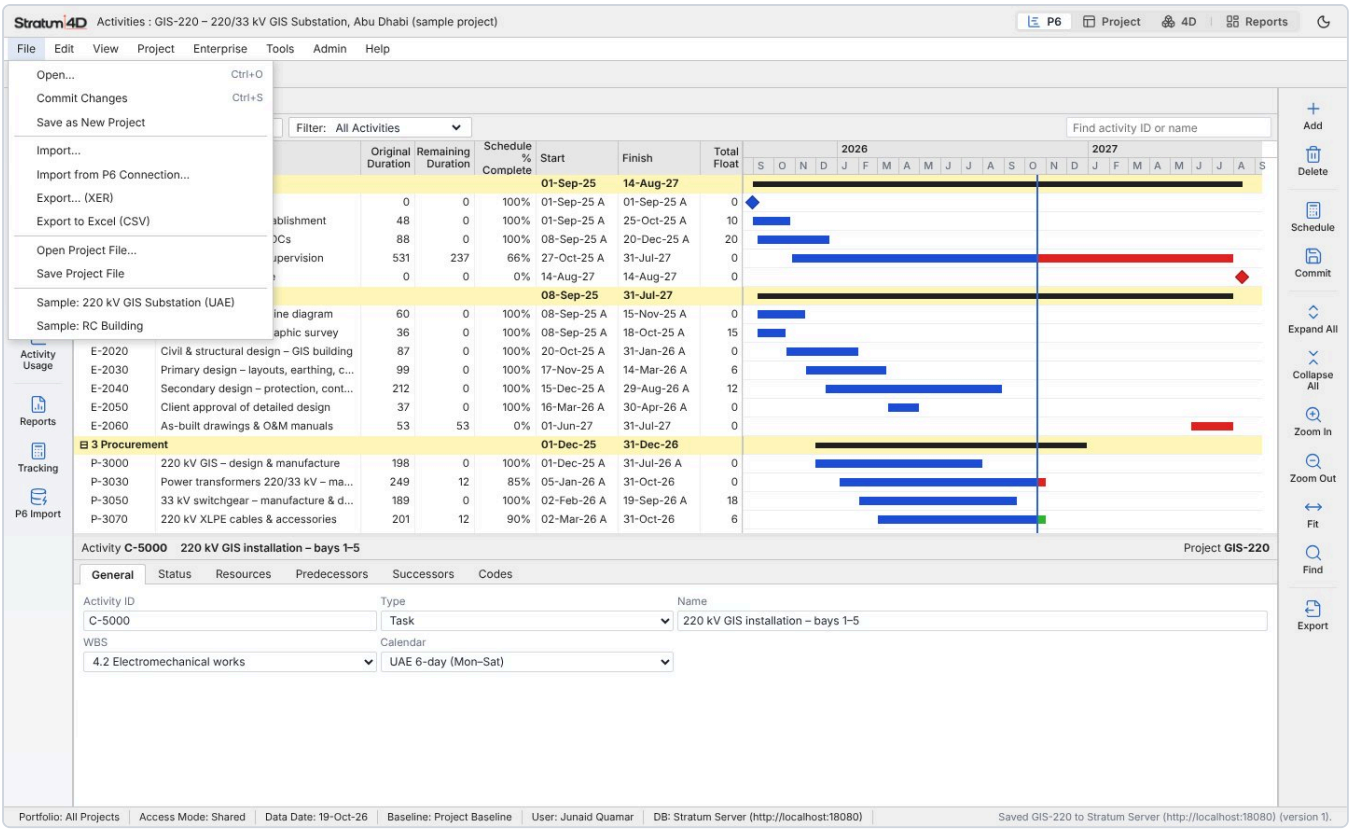


Figure 17. The P6 File menu: open and commit, import (file or P6 connection), export to XER or CSV, project files and the samples.

Export (XER) writes a file that updates the original project when imported in P6 with *Update Existing Project*; tables Stratum did not touch pass straight through. *Export to Excel (CSV)* writes the activity table.

6. 4D simulation

A 4D model links each model element to the activity that builds, removes or uses it, then plays the schedule against the model.

Linking the example model

The example model carries the activity ID in each element's property `Pset_Stratum.ActivityID` (for instance, every part of GIS bays 1–5 holds `C-5000`). One rule therefore links all 202 elements: *element property Pset_Stratum.ActivityID equals Activity ID*.

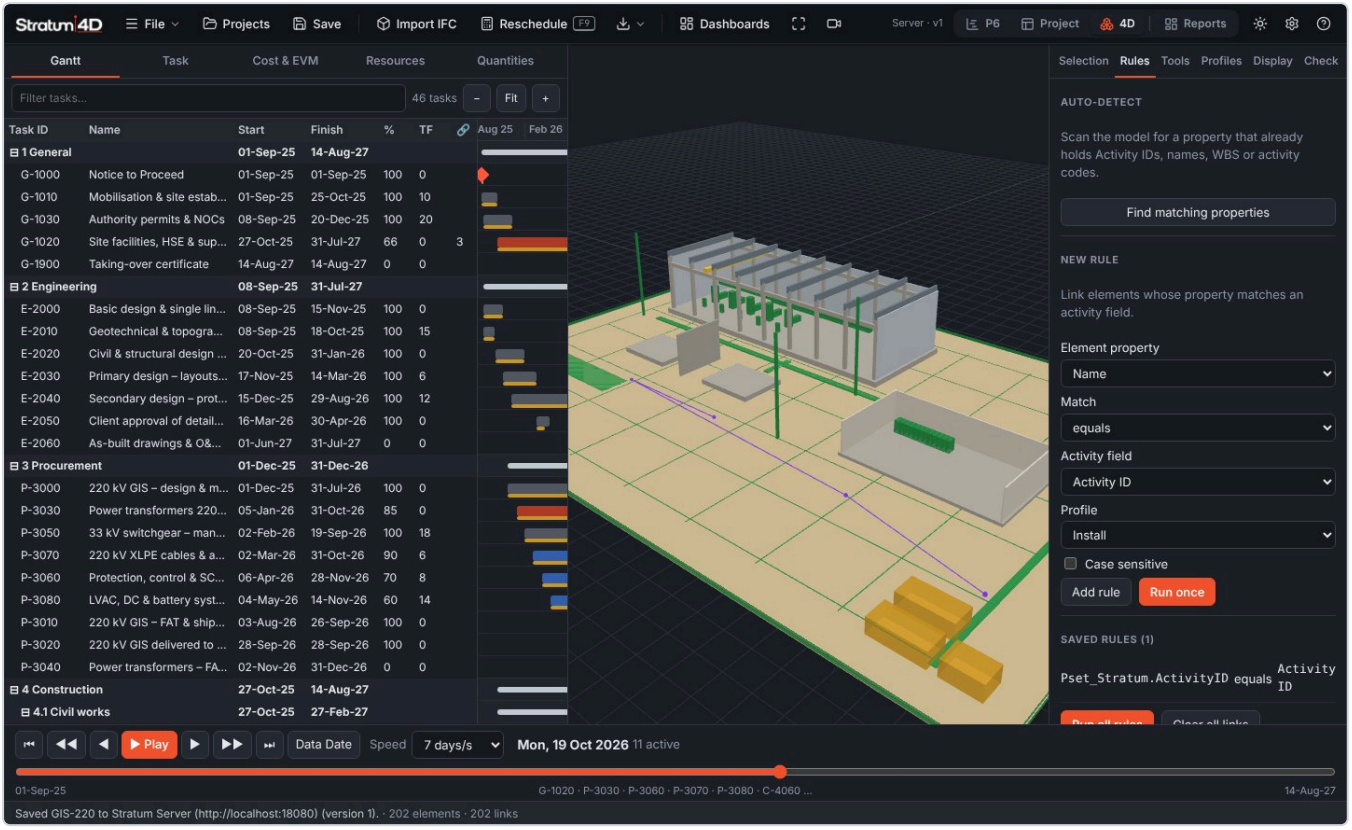


Figure 18. Rules tab. Find matching properties scans the model for a property that already holds activity IDs, names, WBS or codes. Below it, a rule is built from an element property, a match type, an activity field and a profile.

For your own models, the same approach works with a shared parameter from Revit, ArchiCAD or Tekla. Without one, select elements in 3D (or *Selection → Select by property*), pick the activity in the Gantt, choose a profile and press *Link*.

Appearance profiles

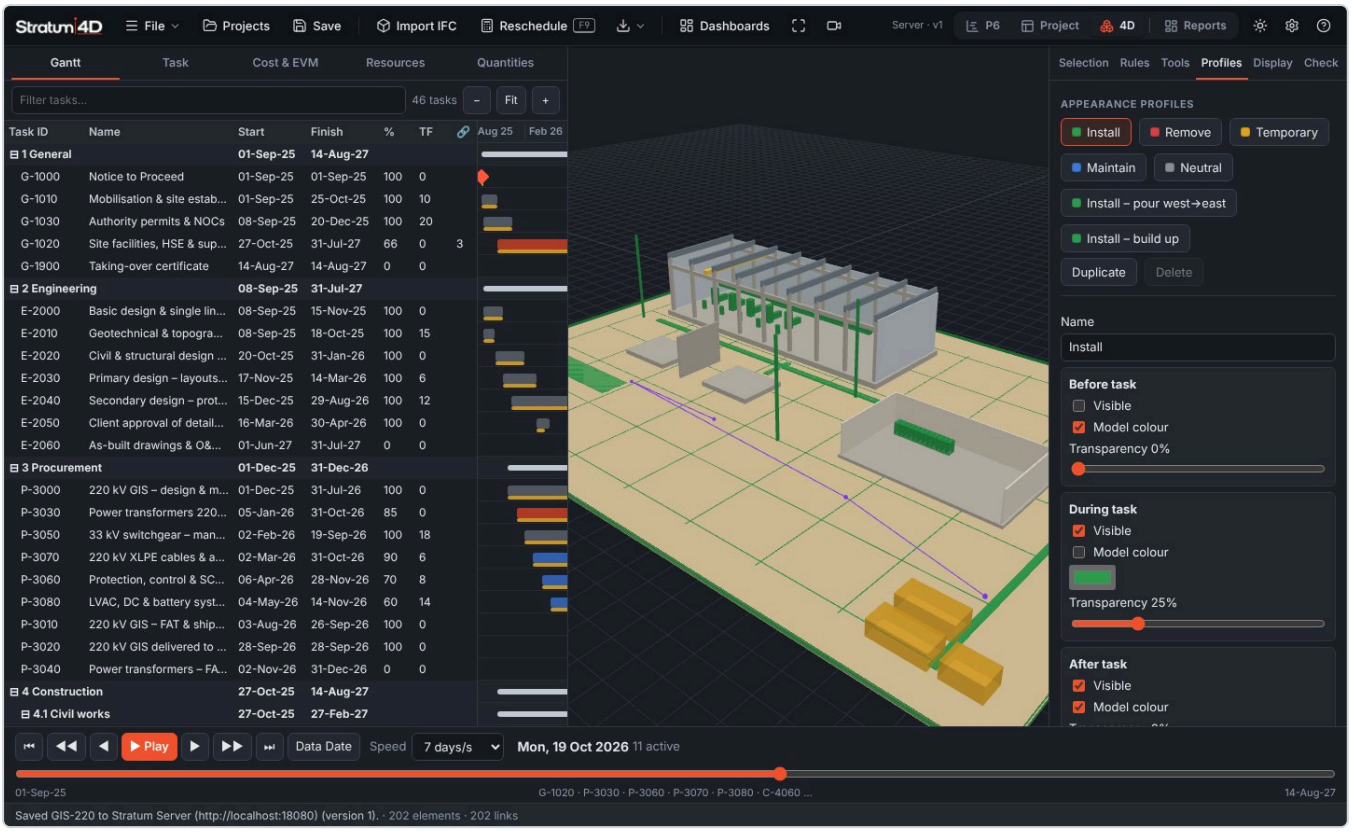


Figure 19. Profiles tab. Each profile sets visibility, colour and transparency before, during and after its activity.

The example uses *Install* for equipment, *Install – pour west→east* for rafts, slabs and trenches, *Install – build up* for walls, columns, piles and cladding, and *Temporary* for the site cabins and the 200 t mobile crane, which appear only while their activity runs.

Groups, pours and equipment paths

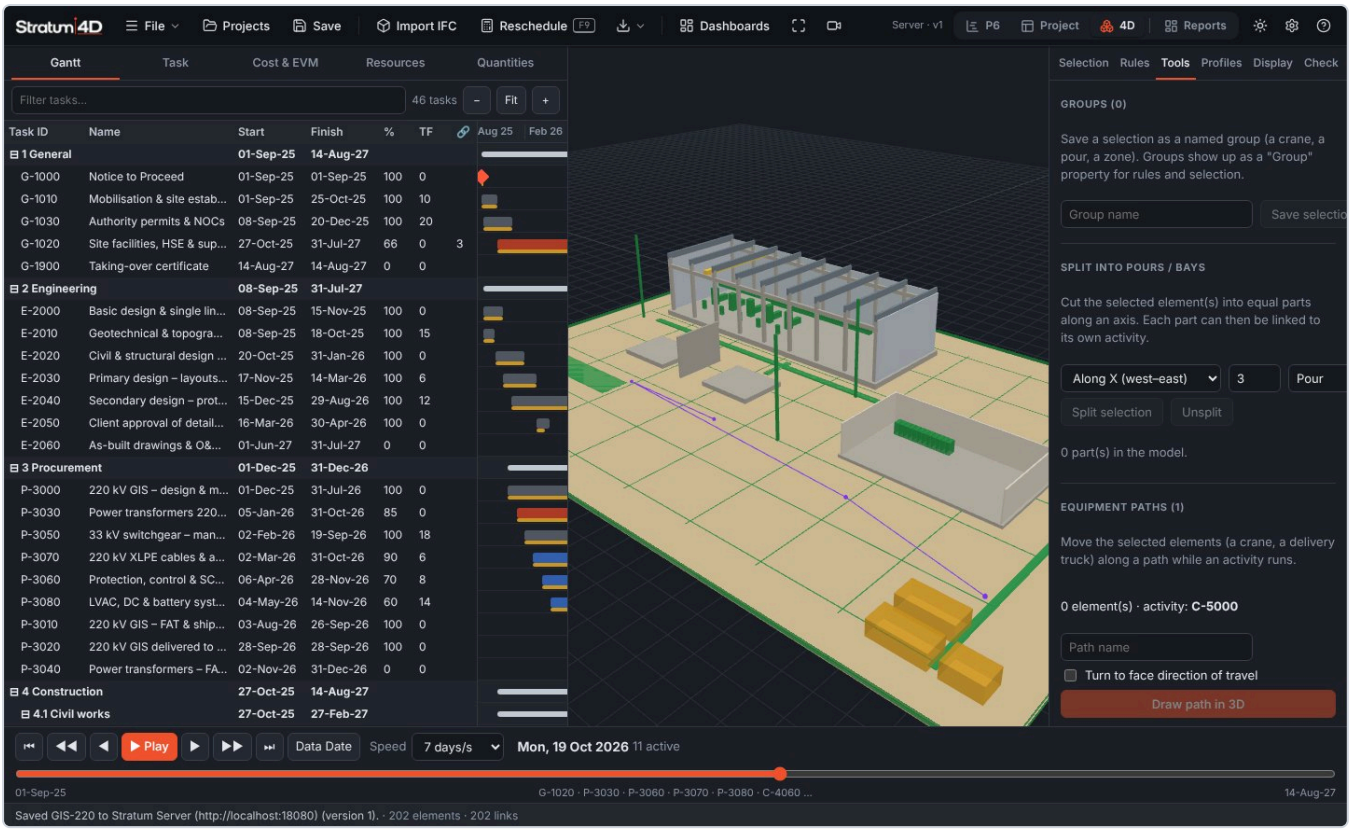


Figure 20. Tools tab: save selections as named groups, split elements into pours or bays, and move equipment along a path.

The example's crane path takes the 200 t mobile crane from the gate along the access road to transformer T1 and then T2 during activity C-4900 (transformer placing).

Display options and checks

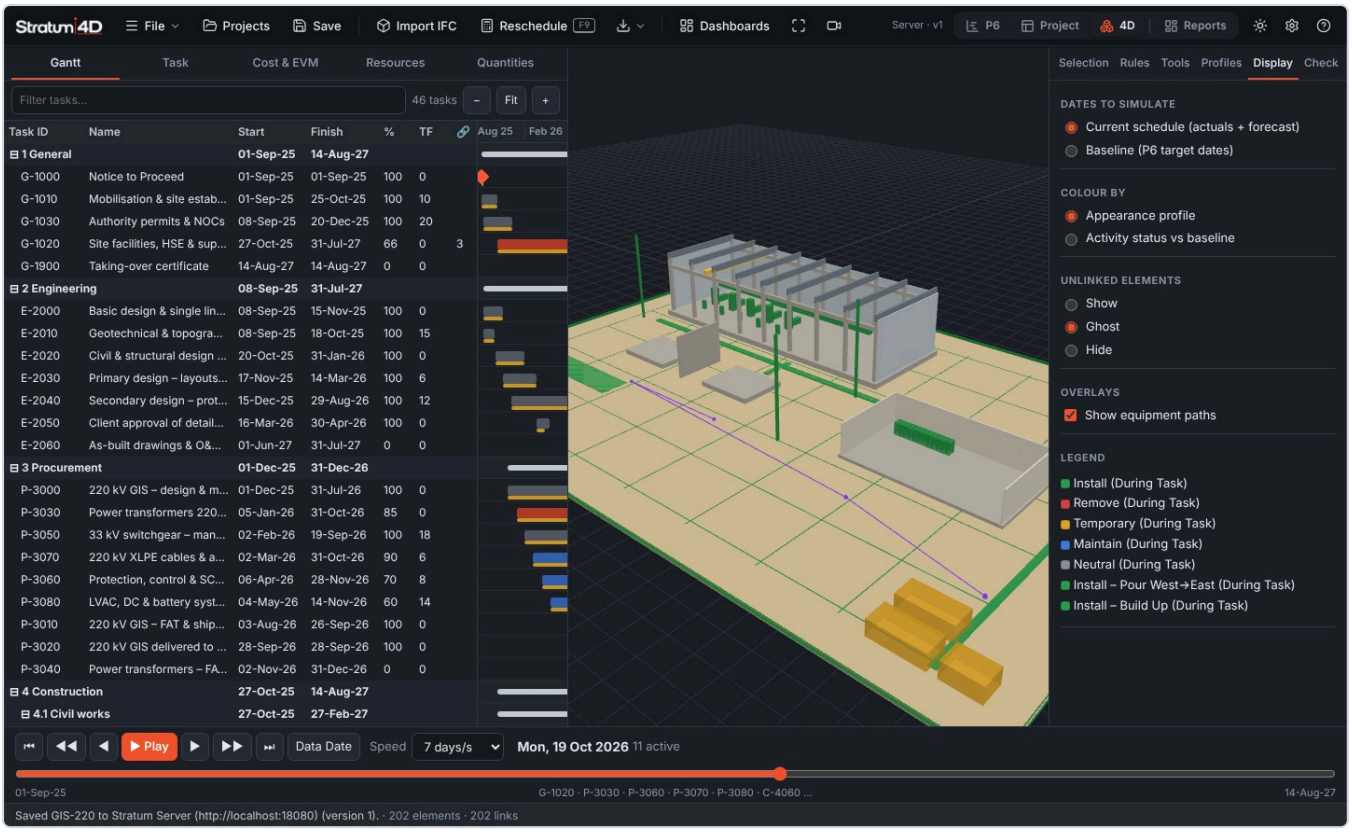


Figure 21. Display tab: simulate current dates or the P6 baseline, colour by profile or by status against baseline, ghost or hide unlinked elements, and show equipment paths.

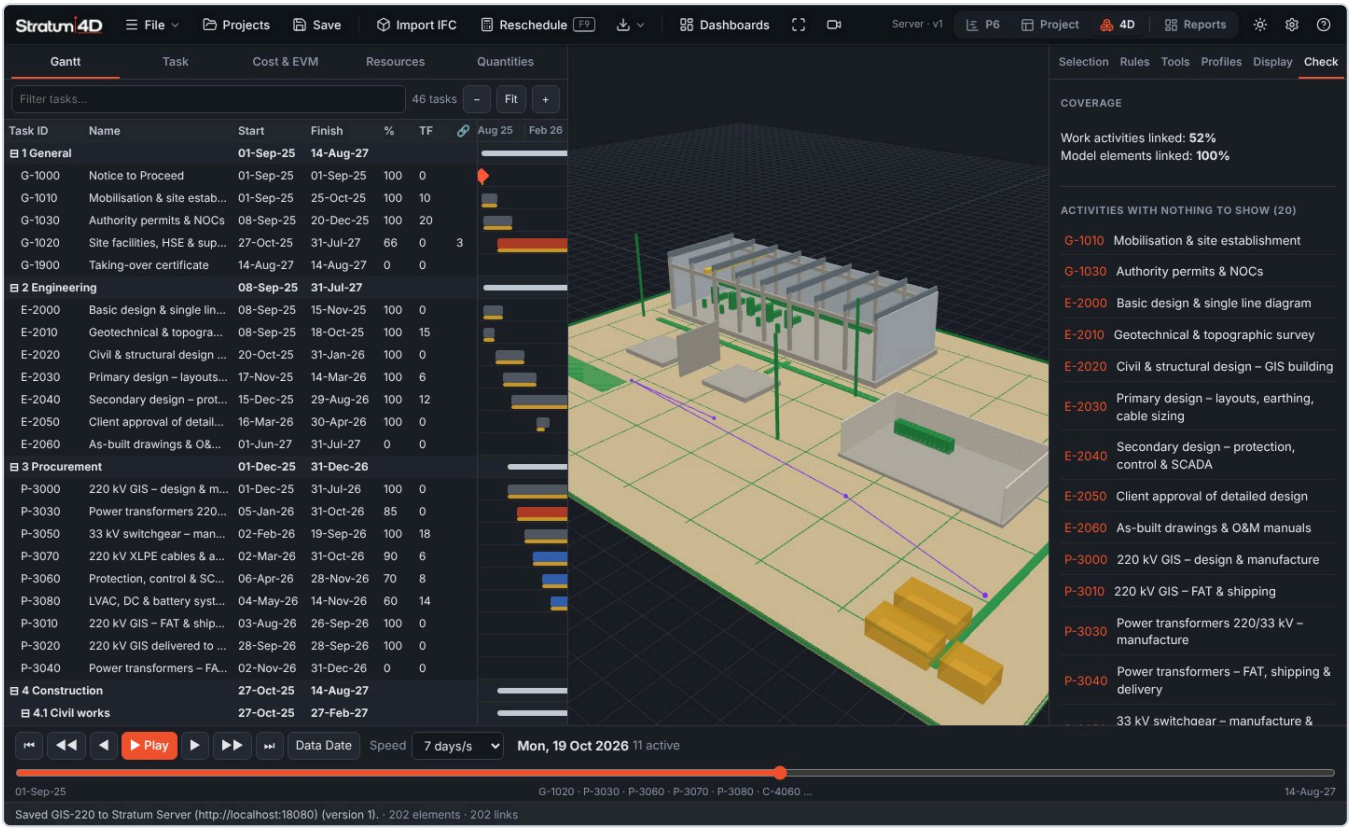


Figure 22. Check tab: link coverage, activities with nothing to show, elements never shown, and the change report after a schedule update.

Playing the sequence

Use the timeline at the bottom: Play/Pause (`Space`), step one day (`←` `→`) or one week (`Shift`+arrows), jump to the start, data date or finish, and set the speed. *Record* saves the playback as a WebM video.

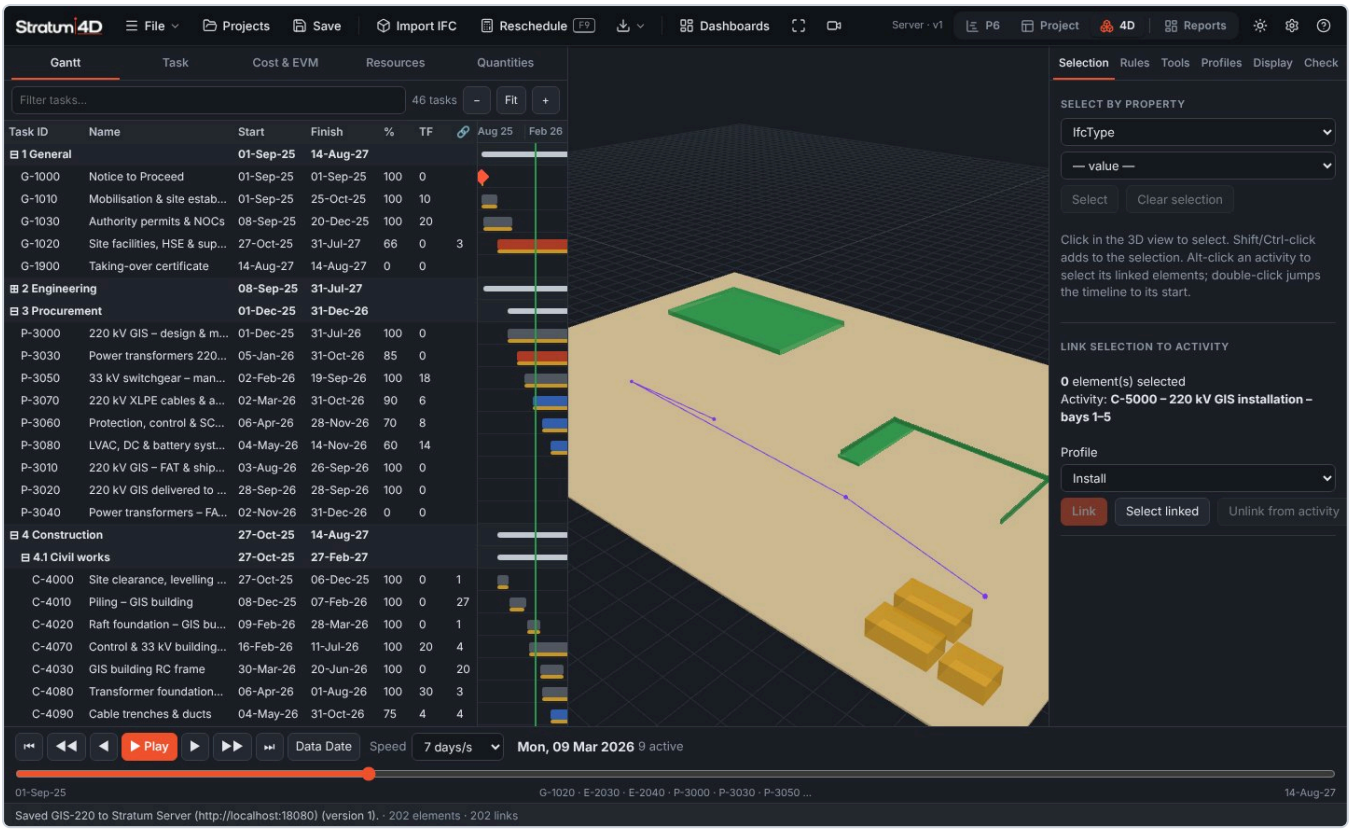


Figure 23. 9 March 2026: site levelled, piles done, the GIS building raft being poured (green) and the control building foundations started. The amber boxes are the site cabins.

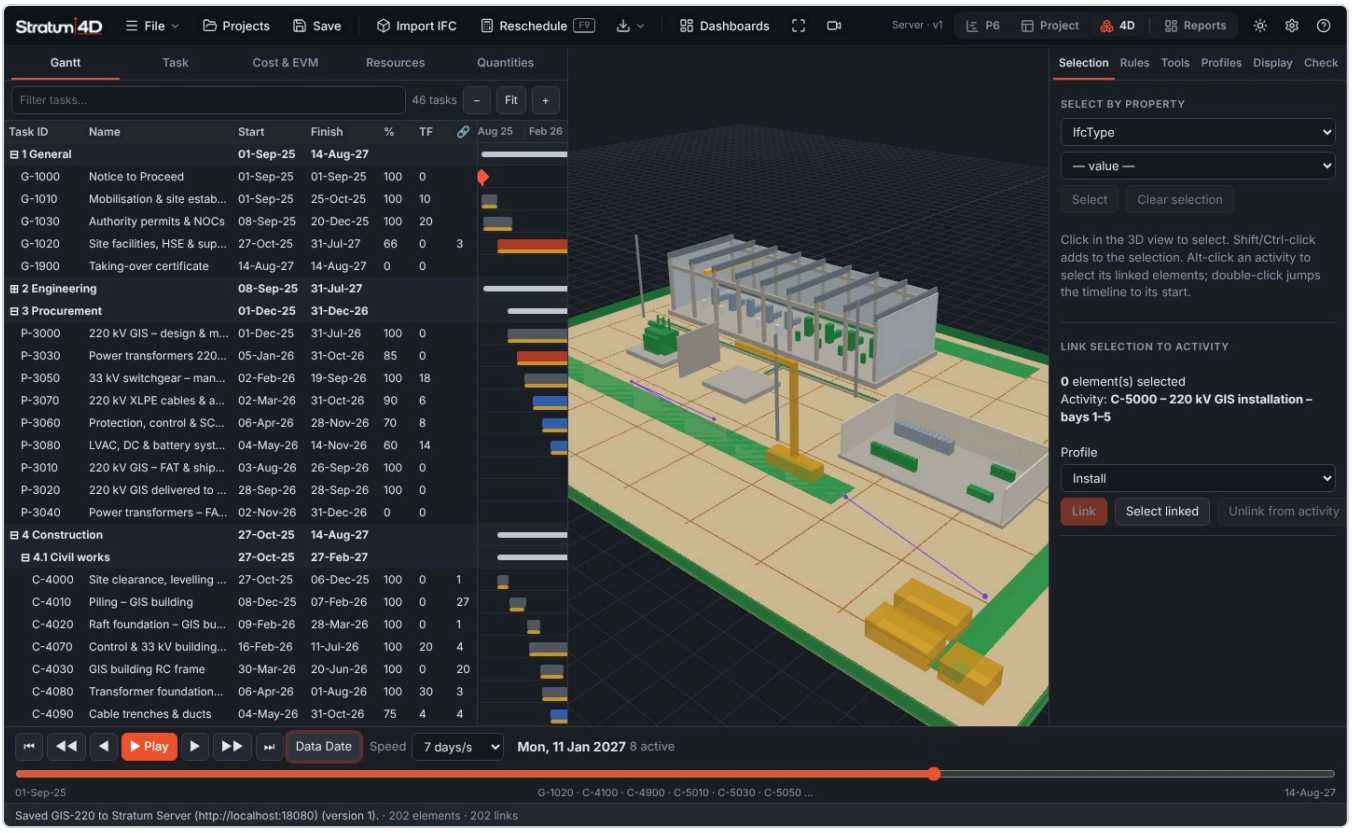


Figure 24. 11 January 2027: GIS hall complete with bays being installed, transformer T1 being erected, the crane (yellow) on its path, cables going into the trenches and the boundary wall under way.

7. Cost, resources and quantities

In the P6 layout, choose what the lower pane shows from the Directory bar or *View → Show on Bottom*: Activity Details, the **Resource Usage Profile** or the **Activity Usage Profile**. In the MS Project layout, Resource Usage is a view of its own (Resource or View tab). Quantities come from the 3D model, so they live in the 4D layout.

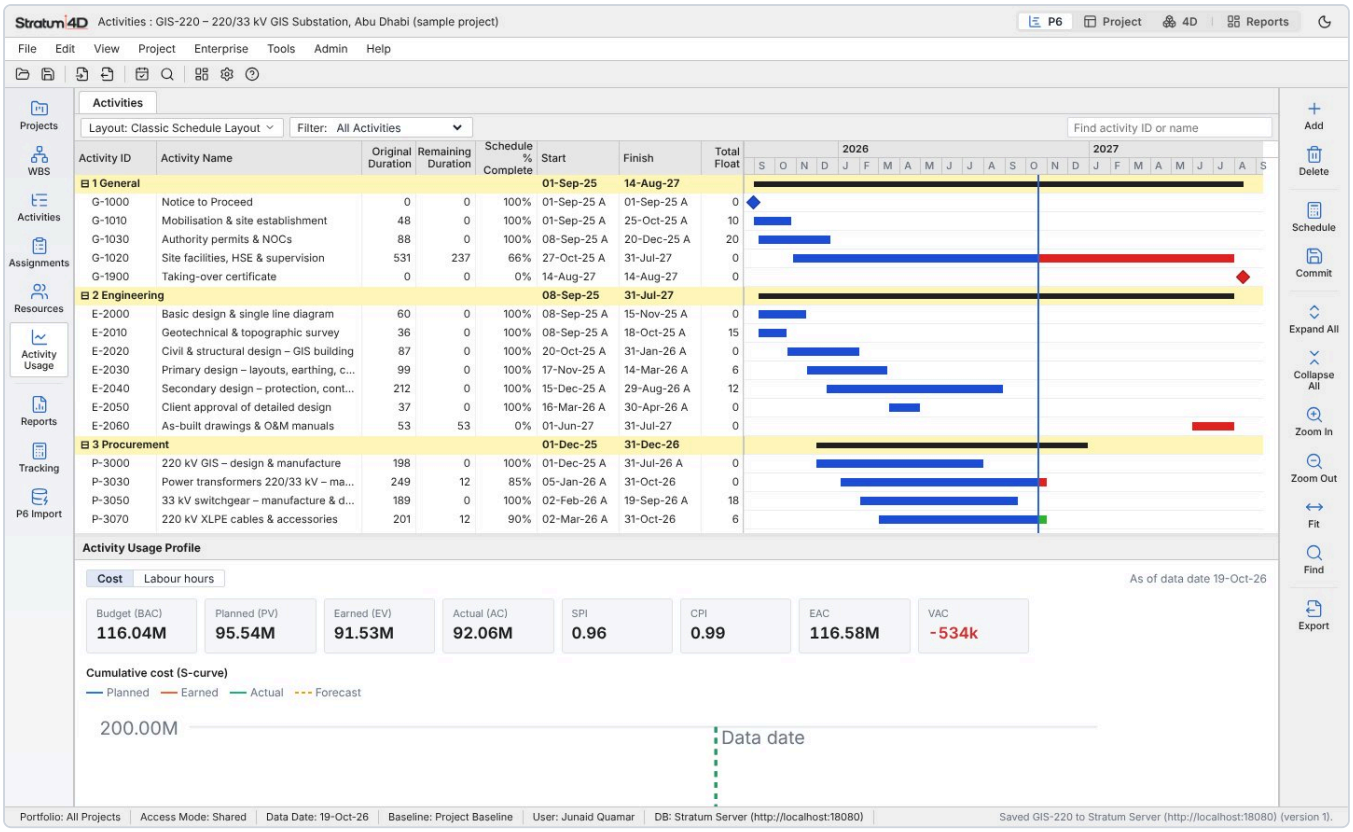


Figure 25. Activity Usage Profile for the example at the data date: budget at completion AED 116.04M, planned value 95.54M, earned value 91.53M, actual cost 92.06M, SPI 0.96, CPI 0.99, EAC 116.58M and VAC −534k, with the cumulative S-curve below.

Earned value is calculated from resource assignments, in cost or labour hours. In the example, SPI 0.96 reflects the late GIS delivery and transformer manufacture; the small cost overrun comes from piling, cladding and secondary design.

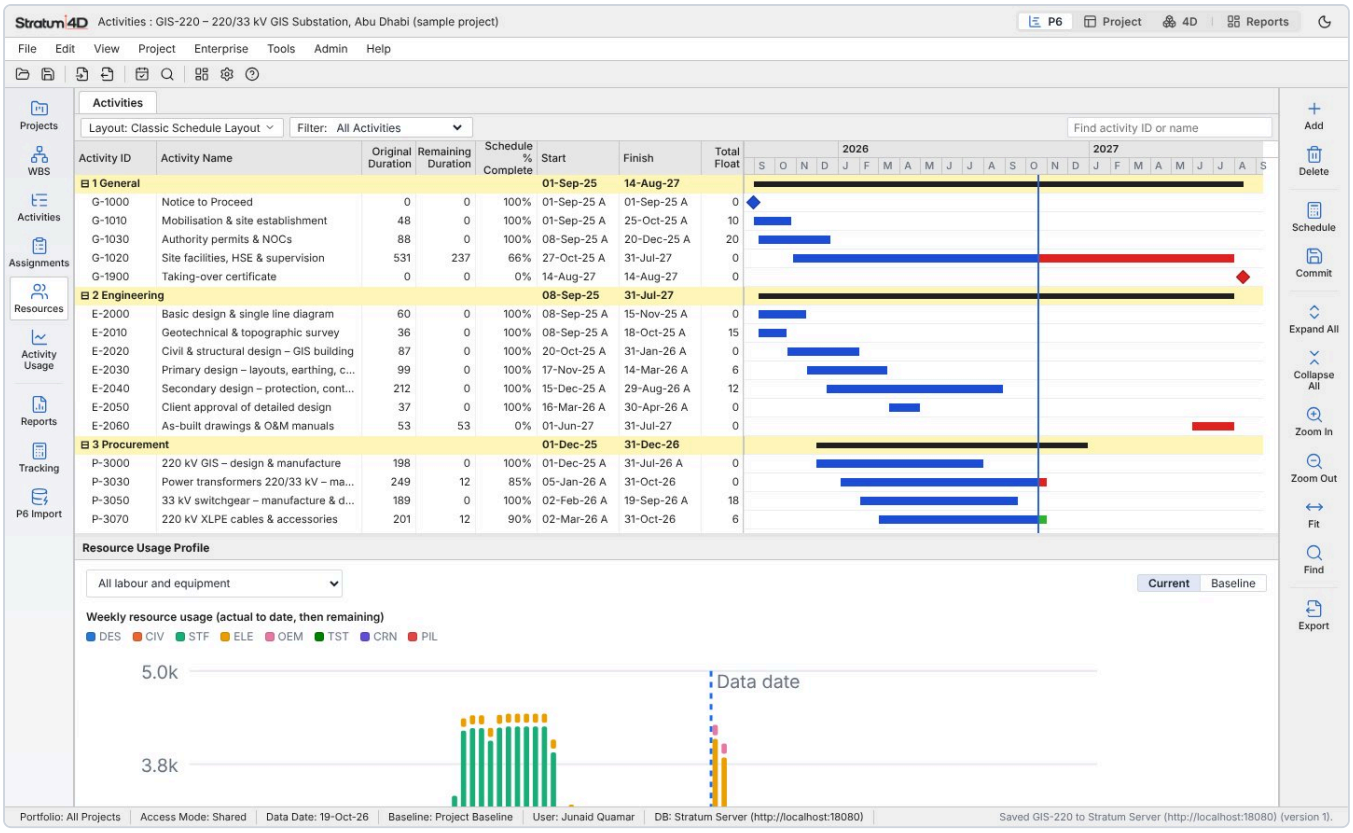


Figure 26. Resource Usage Profile: weekly usage per resource, actual to date then remaining (or the baseline), with a resource table underneath.

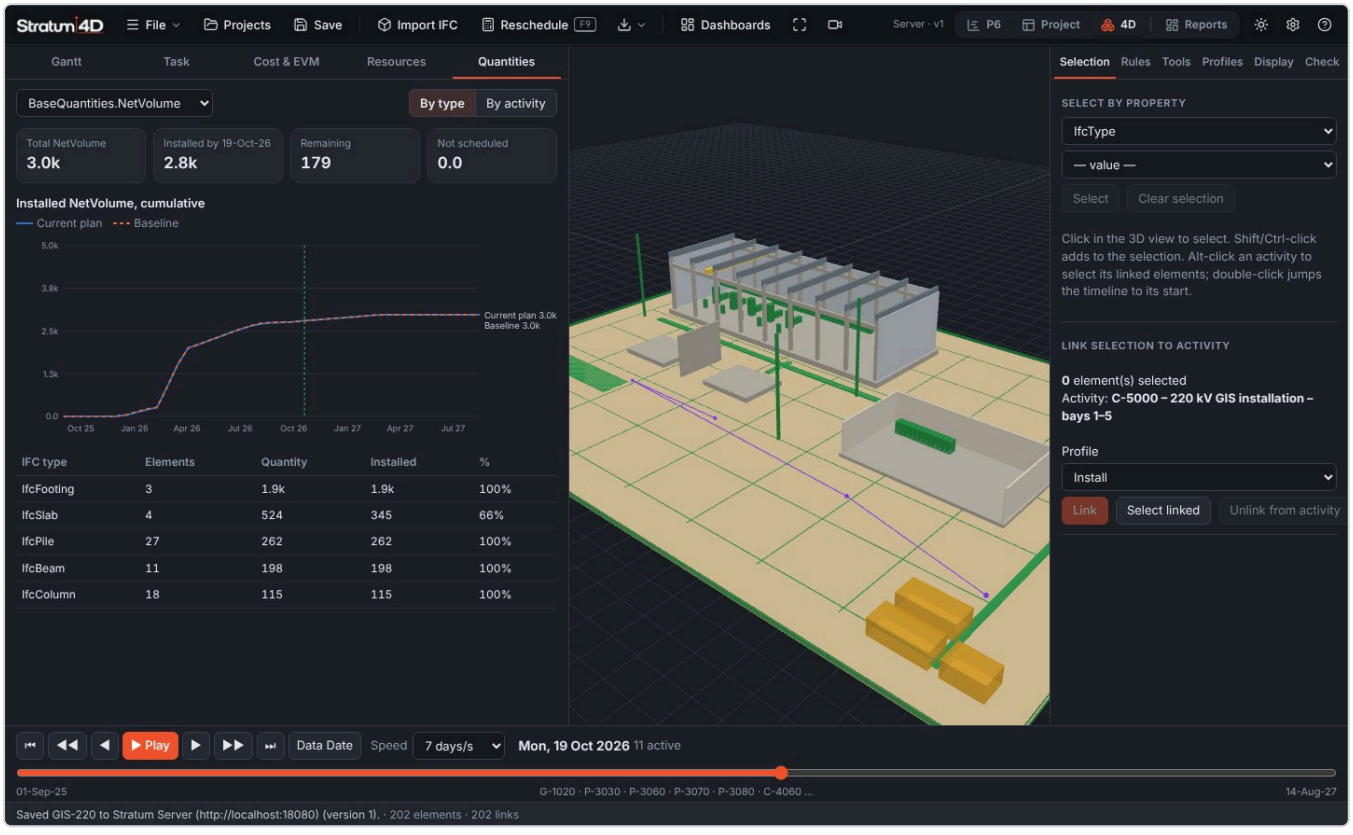


Figure 27. 4D layout → Quantities: any numeric IFC quantity (here NetVolume) installed by the playback date, as tiles and a cumulative curve against the baseline.

8. The project database

Save (*Commit Changes* in the P6 layout, `Ctrl` + `S`) stores the open project in the database: schedule, model links, rules, profiles, groups, pours and paths. Every save gets a version number, shown in the top bar. If someone else saved the project after you opened it, Stratum refuses to overwrite their work and offers *Save as new* instead.

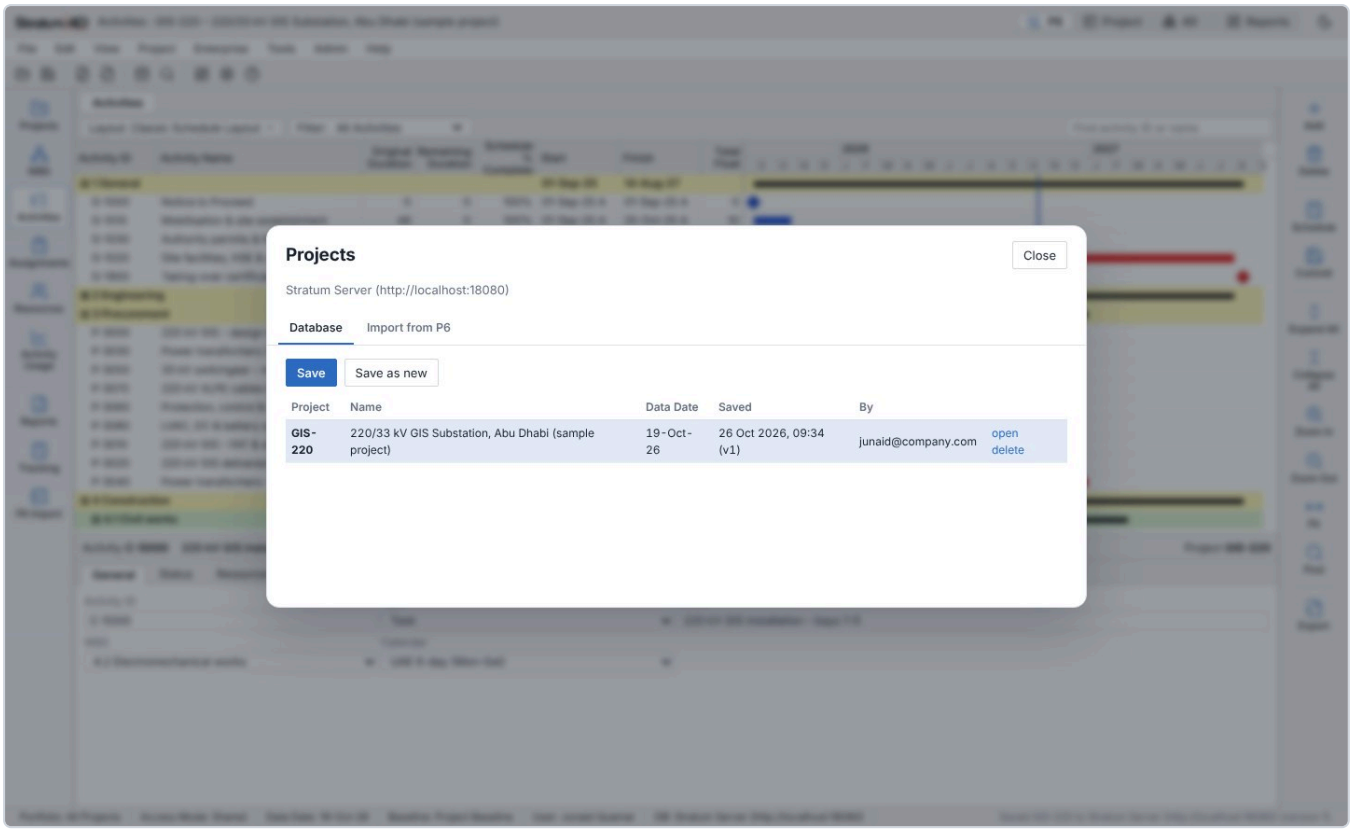


Figure 28. Projects → Database: open, save, save as new or delete. The current project is highlighted with its version.

The model geometry is not stored in the database; open the IFC again after opening a project and the links match by IFC GlobalId. `.stratum.json` project files are still available from the File menu for sharing outside the database.

9. Primavera P6 connections

The desktop app and Stratum Server can talk to P6 directly. The browser cannot, so use XER files there. Connections are set up once in *Settings → Primavera P6*, usually by an administrator.

Connection type	Reads	Writes back	Notes
P6 EPPM (REST API)	Projects, WBS, activities, relationships, calendars, codes, UDFs, resources, assignments	Actual start/finish, % complete, remaining duration	Address normally ends in /p6ws/restapi. Username/password or a bearer token (Oracle-hosted P6).
P6 Professional database (SQL Server, Oracle, SQLite)	The same tables as an XER export	No (read-only by design)	Use a read-only database login; send changes back by XER.

Checking a connection

Test connection checks each link in the chain and stops at the first one that breaks, with a plain-language fix. *Copy report for IT* puts the result on the clipboard.

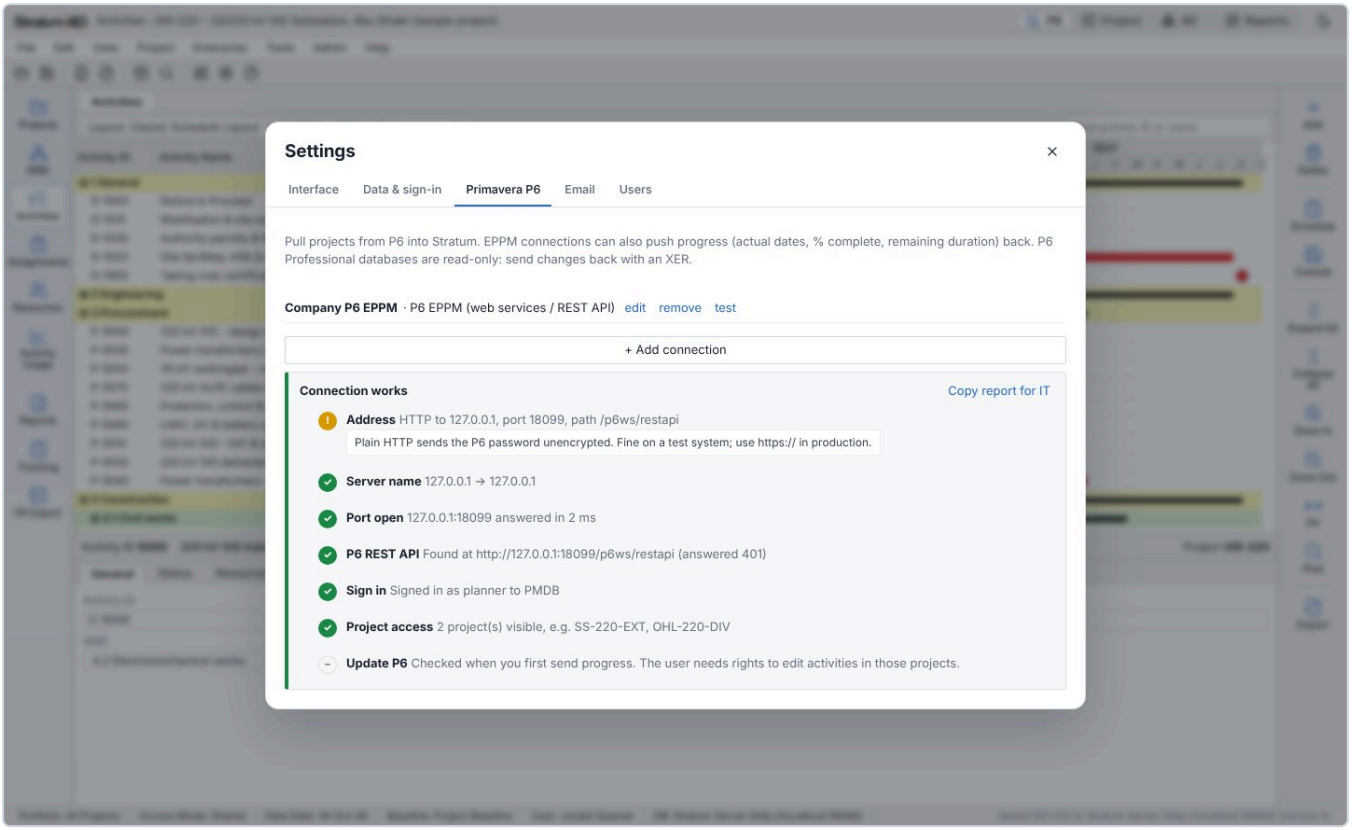


Figure 29. A successful check: address, server name, port, P6 REST API, sign-in and project access.

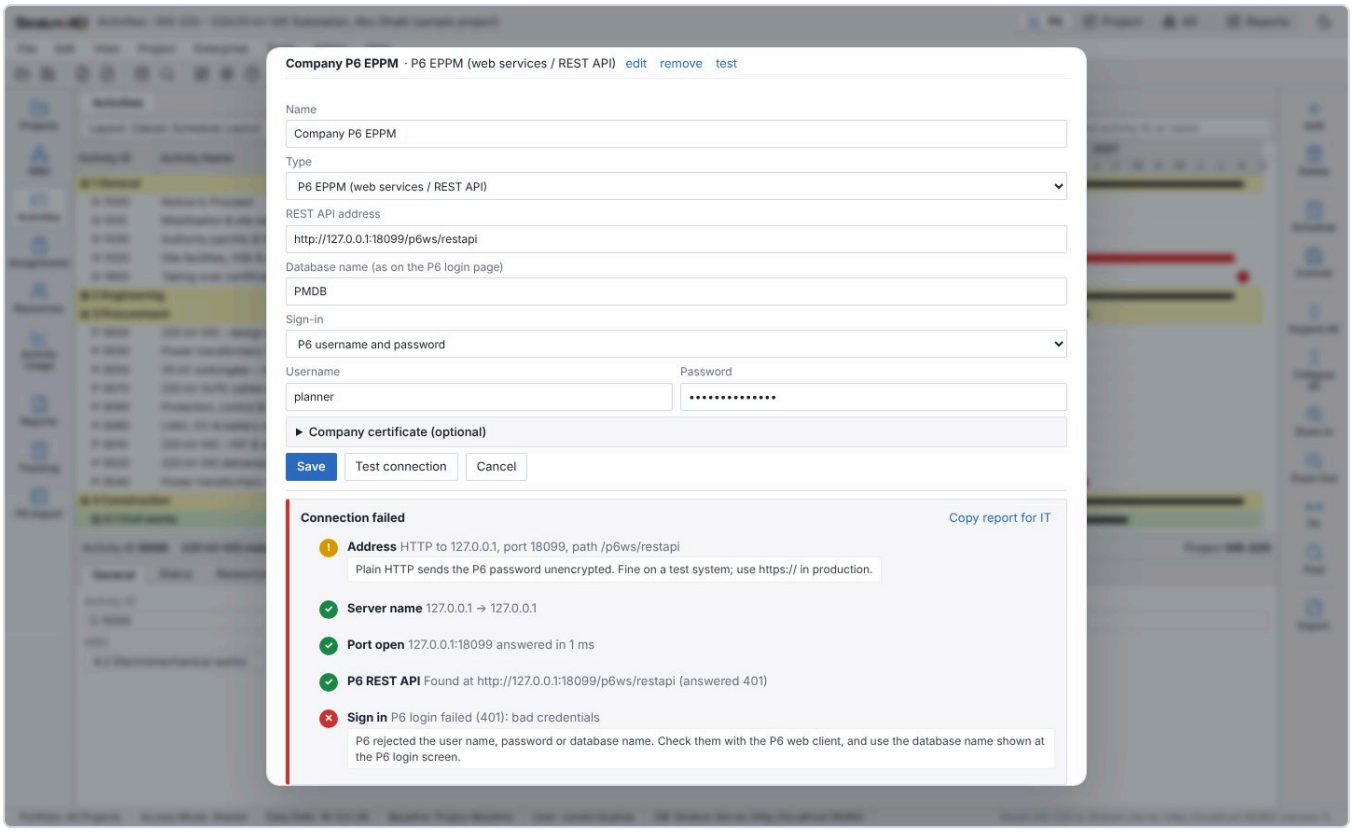


Figure 30. A failed check: sign-in rejected, with what to check.

Step	What it catches
Address	Not a web address; plain http:// (password sent unencrypted); path not ending in /restapi
Server name	Name not found: spelling, or not on the company network / VPN
Port open	Connection refused (P6 stopped, wrong port) or no answer (firewall, VPN)
Certificate	Expired, issued for a different name, or from a company authority this machine does not trust
P6 REST API	Nothing at this path (REST not deployed), or an application-server error
Sign in	Wrong user, password or database name; user without API access; single sign-on
Project access	Signed in but no projects visible: give the user OBS access

Network and certificates

Stratum only makes outgoing HTTPS calls to P6, so nothing needs port forwarding on the Stratum side. The machine running Stratum must reach the P6 server the same way a browser reaches the P6 web client. If Stratum Server runs in the cloud and P6 on company servers, connect them with a VPN or private link rather than exposing P6 to the internet.

- **Desktop app:** P6 calls use the operating system's network stack, so certificates installed by IT in Windows or the macOS keychain are trusted, and the system proxy is used.
- **Stratum Server:** set `NODE_EXTRA_CA_CERTS` to a file holding the company root certificate, and `HTTPS_PROXY` / `NO_PROXY` if a proxy is needed.
- **Either:** paste the root certificate into *Company certificate (optional)* on the connection.

Importing and updating P6

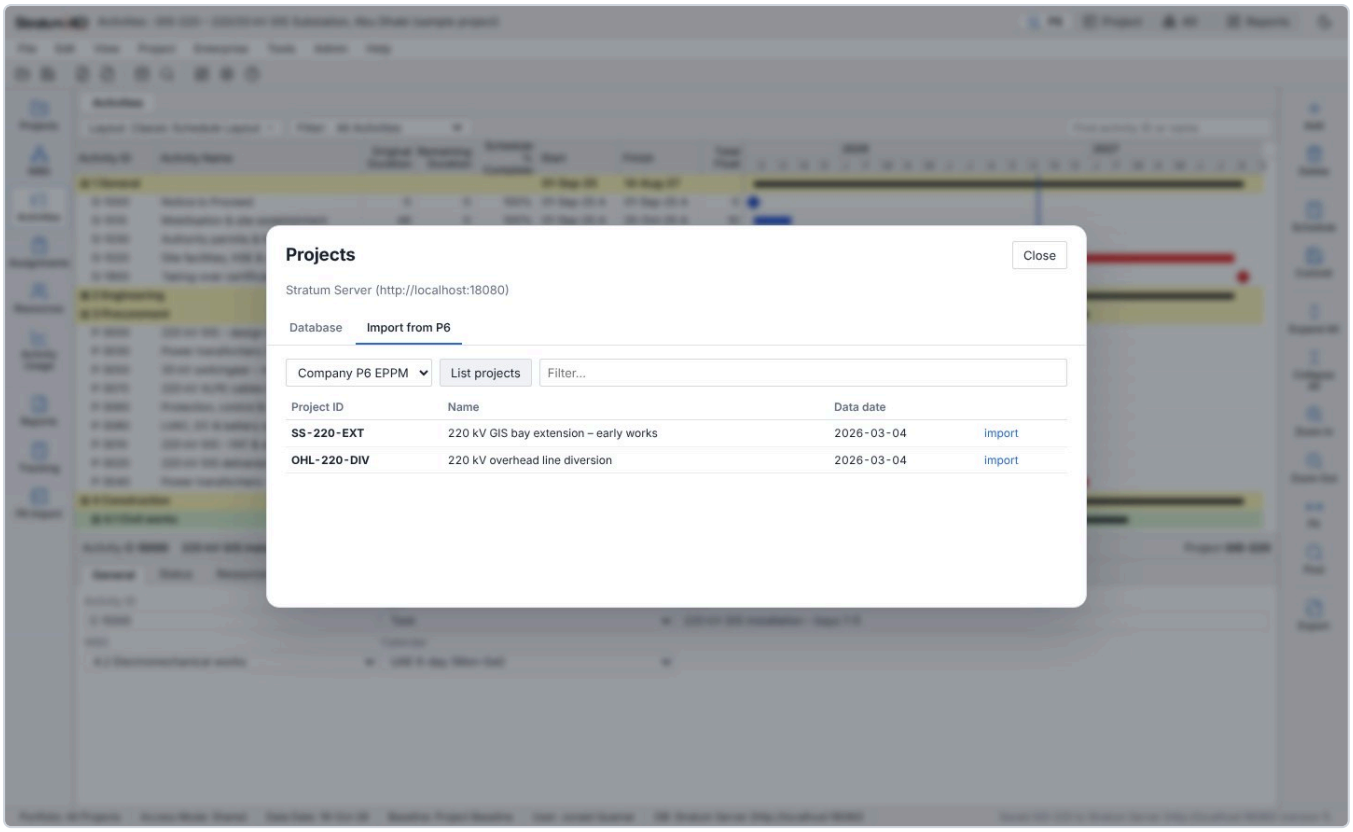


Figure 31. Projects → Import from P6: list the projects on a connection and import one.

An imported project behaves like any other: link it to the model, record progress and report on it. For EPPM projects the top bar shows *Update P6*, which sends actual dates, % complete and remaining duration back for activities that exist in P6. Activities added in Stratum are listed, not pushed. After the update, reschedule in P6 with F9.

10. Dashboards and reports

Reports in the top switch opens the report builder, in the spirit of Power BI. It works the same from the P6, MS Project and 4D layouts. On the left are your dashboards, your own templates and the starting points under *New*. Each dashboard shows the open project or a saved project chosen at the top.

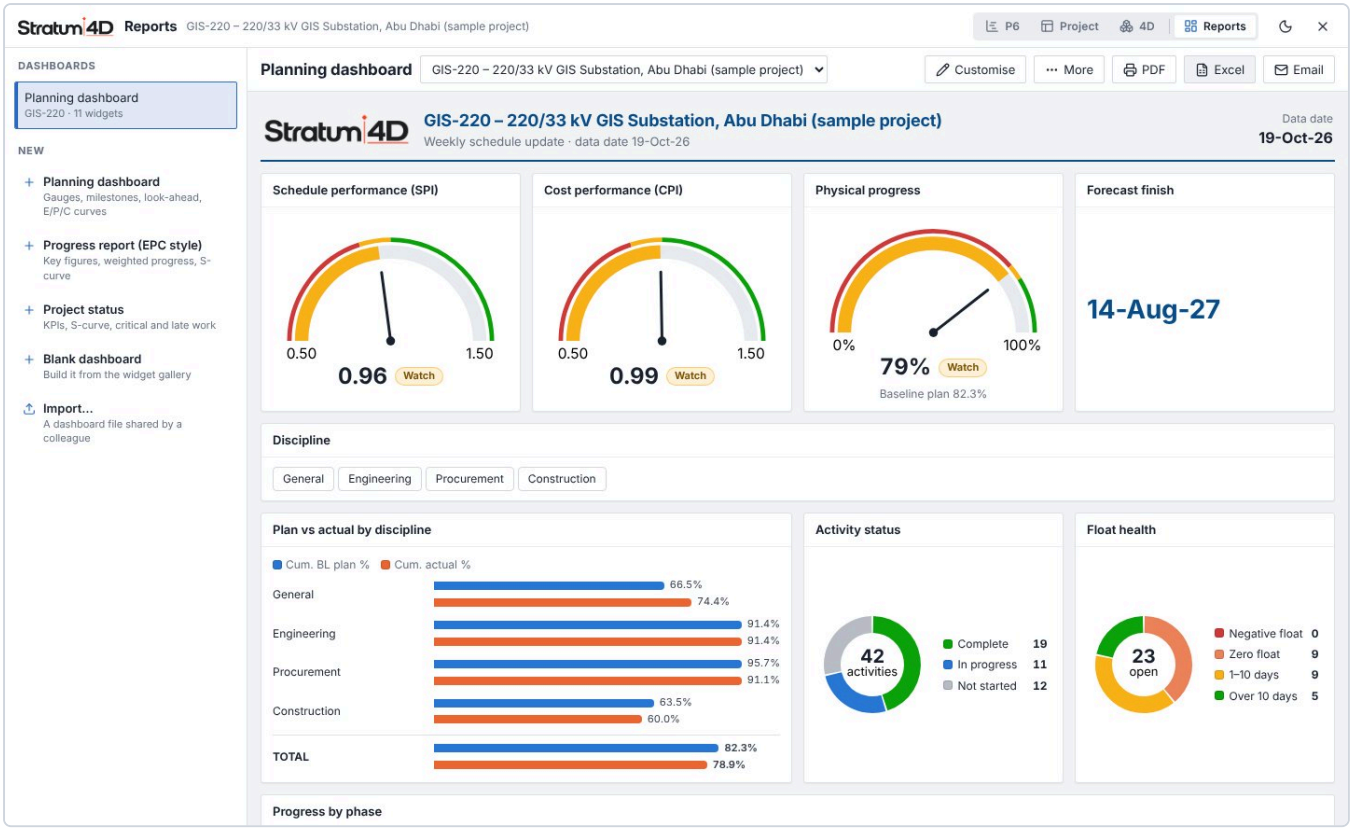


Figure 32. The Planning dashboard template on the example project: SPI and CPI gauges with red, amber and green limits, physical progress against the baseline plan, forecast finish, a Discipline slicer, plan against actual by discipline, activity status and float health.

The Planning dashboard collects the views planners most often post as their weekly P6 dashboards: SPI and CPI gauges, a milestone tracker, plan against actual by discipline, a separate S-curve for each of engineering, procurement and construction, and a look-ahead. In the example, SPI is 0.96 and CPI 0.99, both in the amber band between 0.90 and 1.00.

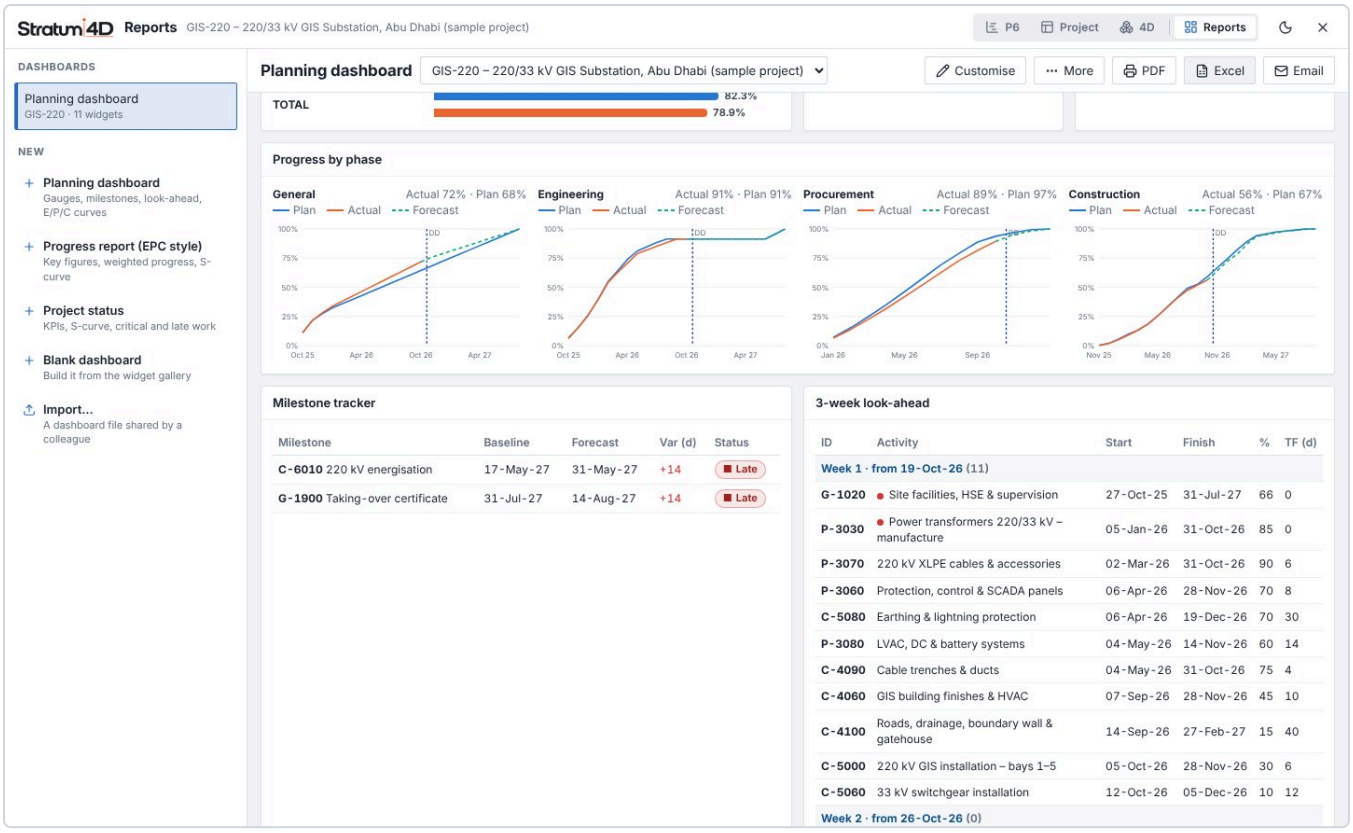


Figure 33. Lower half: one progress S-curve per WBS phase, the milestone tracker (both open milestones forecast 14 days late) and the 3-week look-ahead grouped by week from the data date.

The EPC progress report

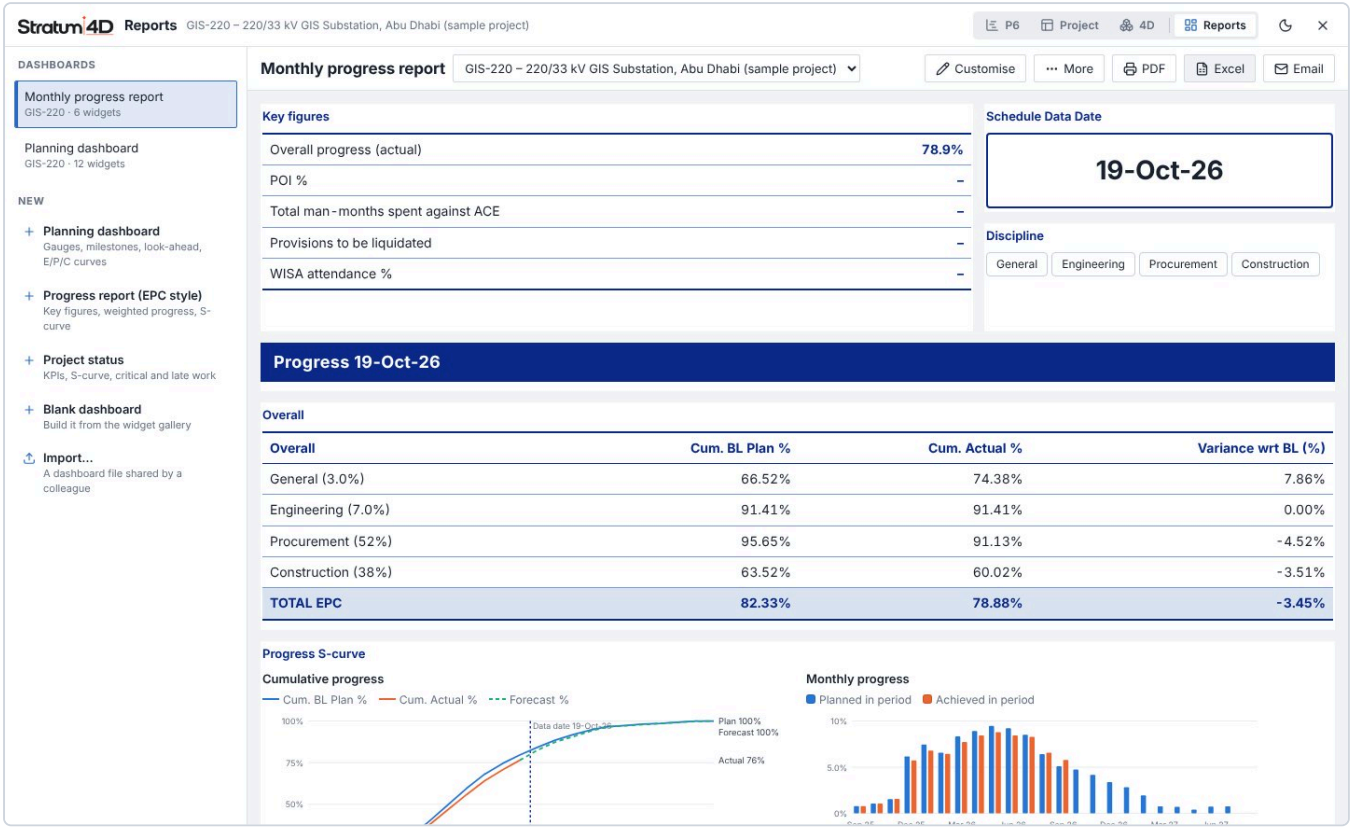


Figure 34. The Progress report (EPC style) template on the example project: key figures, the Schedule Data Date card, the Discipline slicer, the navy Progress banner and the weighted progress table with its TOTAL EPC row.

Progress is weighted by budgeted cost, labour hours, duration or equally, and grouped by WBS level 1 or 2, any activity code or any user field. With the example's cost loading, the weights come out at:

WBS level 1	Weight by cost	Cum. BL plan %	Cum. actual %	Variance
General	3.0%	66.52%	74.38%	7.86%
Engineering	7.0%	91.41%	91.41%	0.00%
Procurement	52.1%	95.65%	91.13%	-4.52%
Construction	37.8%	63.52%	60.02%	-3.51%
TOTAL	100%	**82.33%**	**78.88%**	**-3.45%**

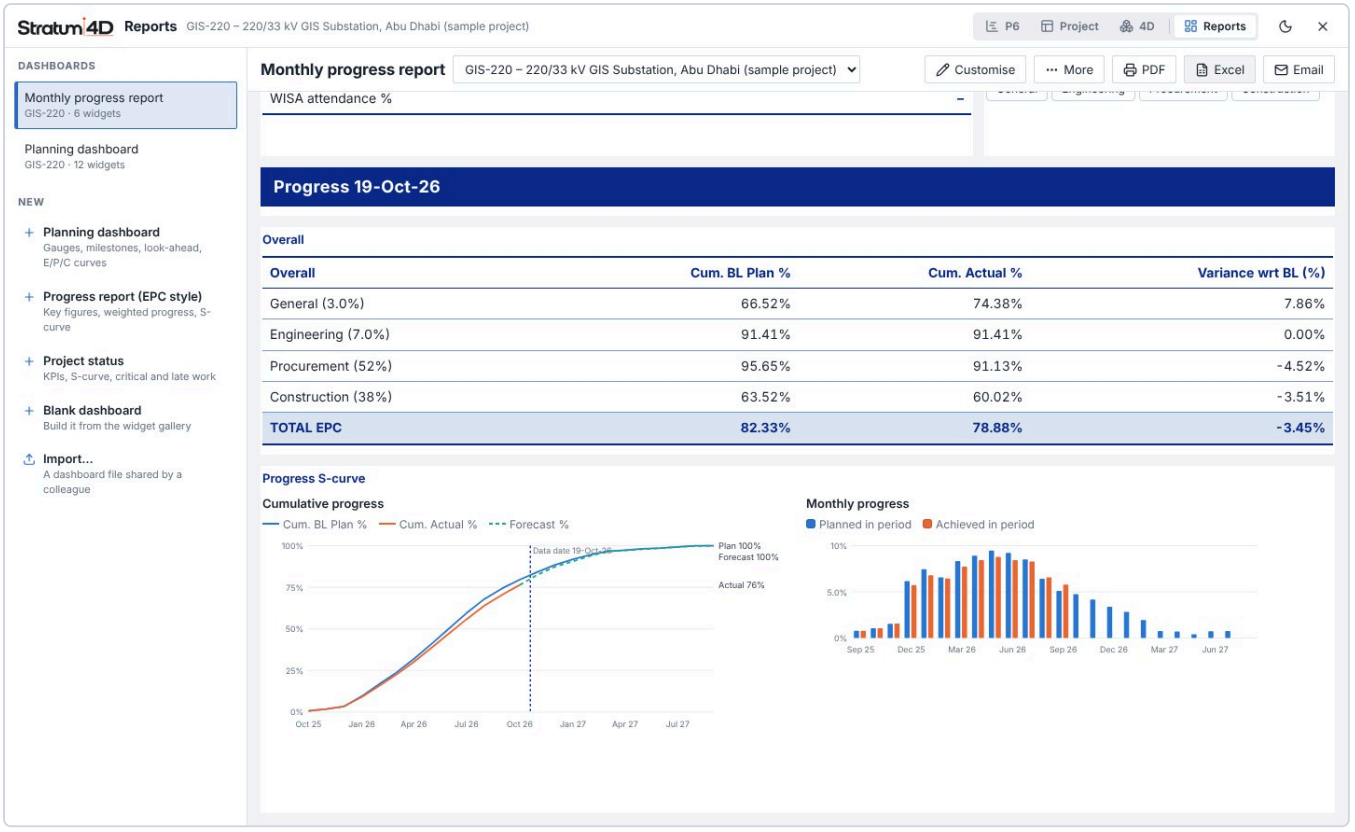


Figure 35. Lower half of the report: cumulative BL plan, actual and forecast S-curve, with monthly planned and achieved progress beside it.

Slicers

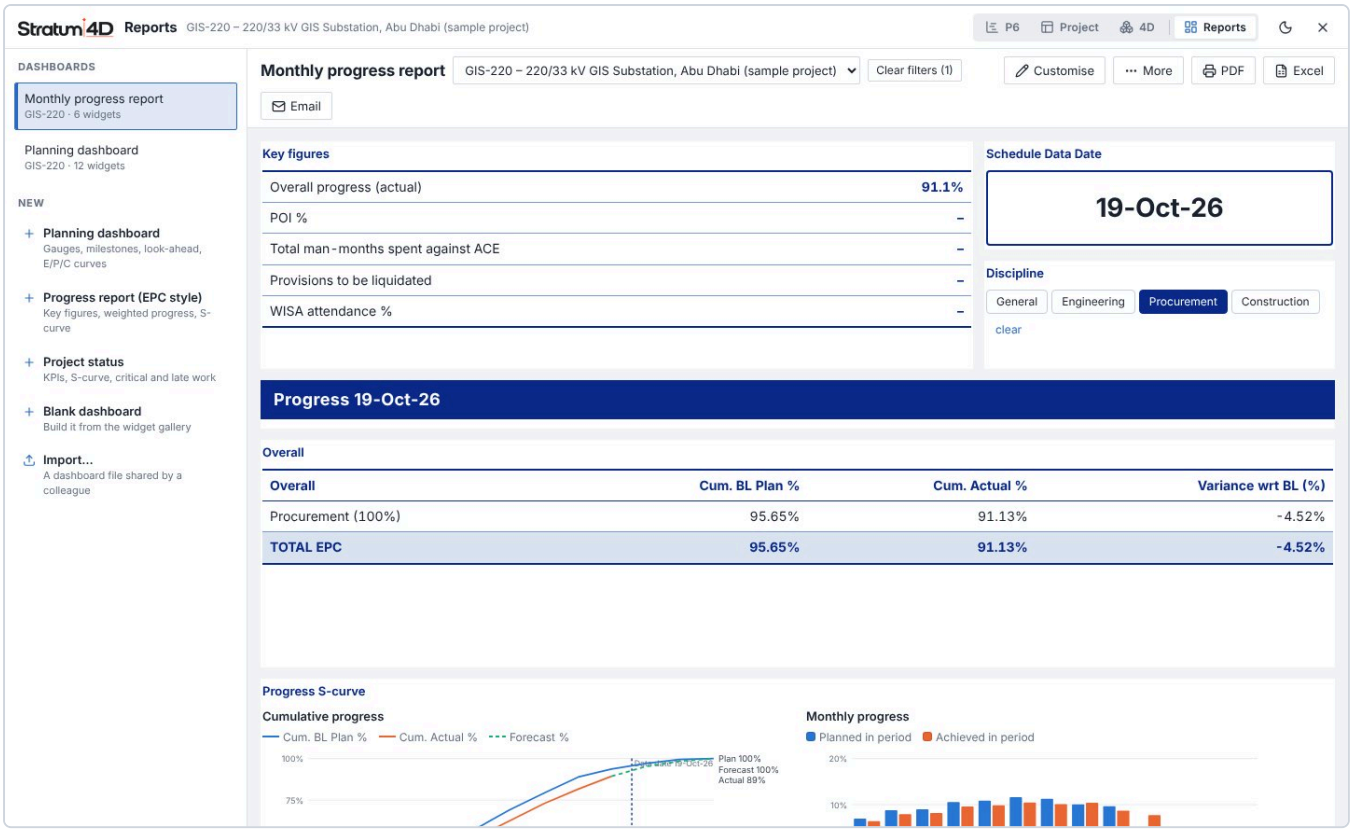


Figure 36. Clicking Procurement in the slicer filters every widget to procurement activities; the TOTAL row recomputes for the selection. Click again, or Clear filters, to remove it.

Customising a dashboard

Click *Customise*. The canvas stays live while you edit, and a panel opens on the right:

- **Move and resize.** Drag a card by its title bar to move it. Drag its bottom-right corner to resize it: widths run from 1/12 to the full page and heights in half-row steps. The size chip on the card shows the result, for example 6×2.
- **Data.** Title, widget type, project, measure, weighting basis, grouping, period and anything else the widget reads. KPI tiles and gauges take decimals, a prefix or suffix (such as `AED `), and the values at which they turn green and red.
- **Look.** Accent colour, card background, title bar (plain, underlined or filled), card style, text size, alignment, line or shaded-area charts, legend on or off, and which series to show.
- **Filter.** Limit one widget to part of the project, for example only Construction, and choose whether it follows the dashboard's slicer buttons.
- **Tables.** Choose columns (ID, name, WBS, status, %, current, baseline and actual dates, variance, total float, original and remaining duration, critical, any activity code), put them in order and set the default sort. Readers can click a heading to sort.

`Ctrl` + `Z` and `Ctrl` + `Y` undo and redo; `Del` removes the selected widget; the copy icon duplicates it.

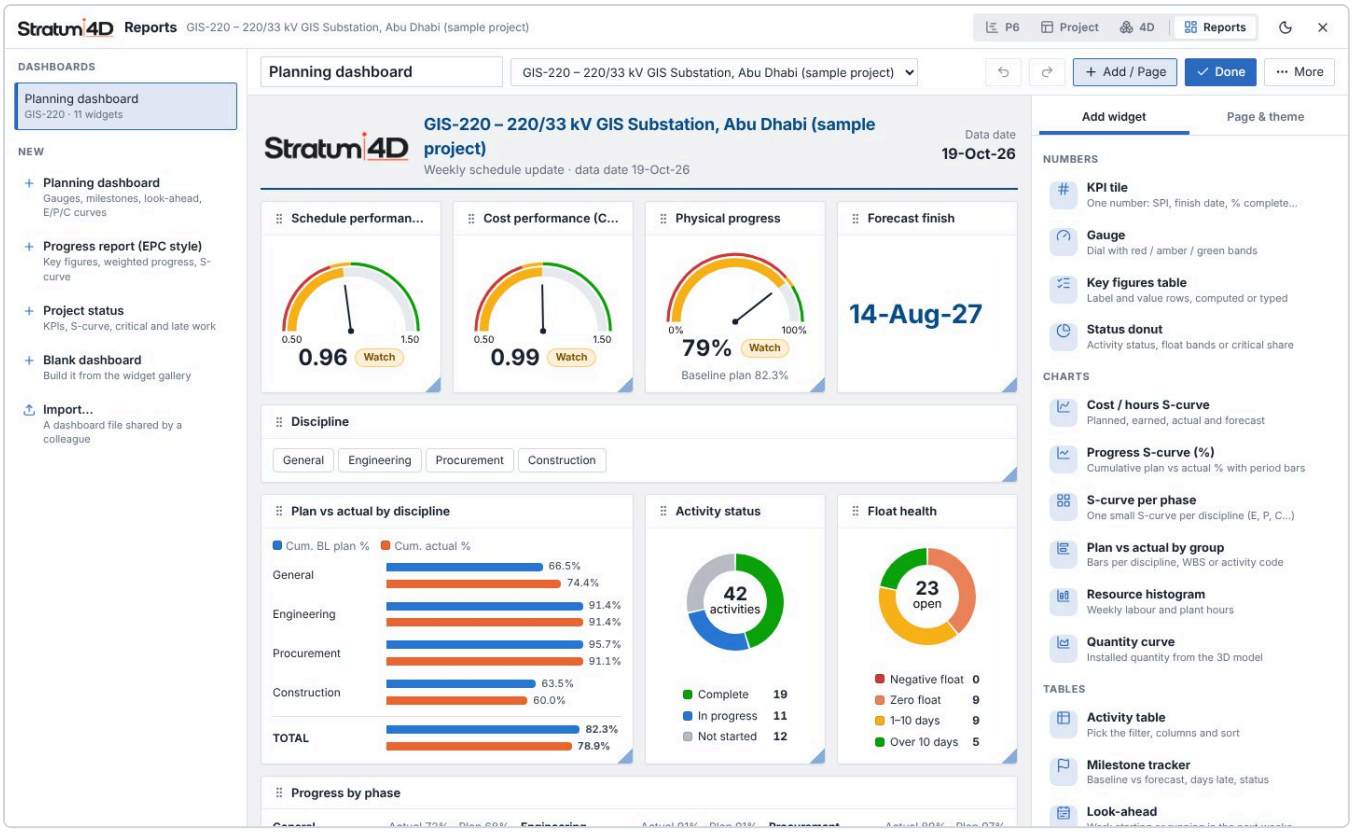


Figure 37. Customise with nothing selected: the widget gallery, grouped into numbers, charts, tables and layout. Click a widget to add it at the end of the page.

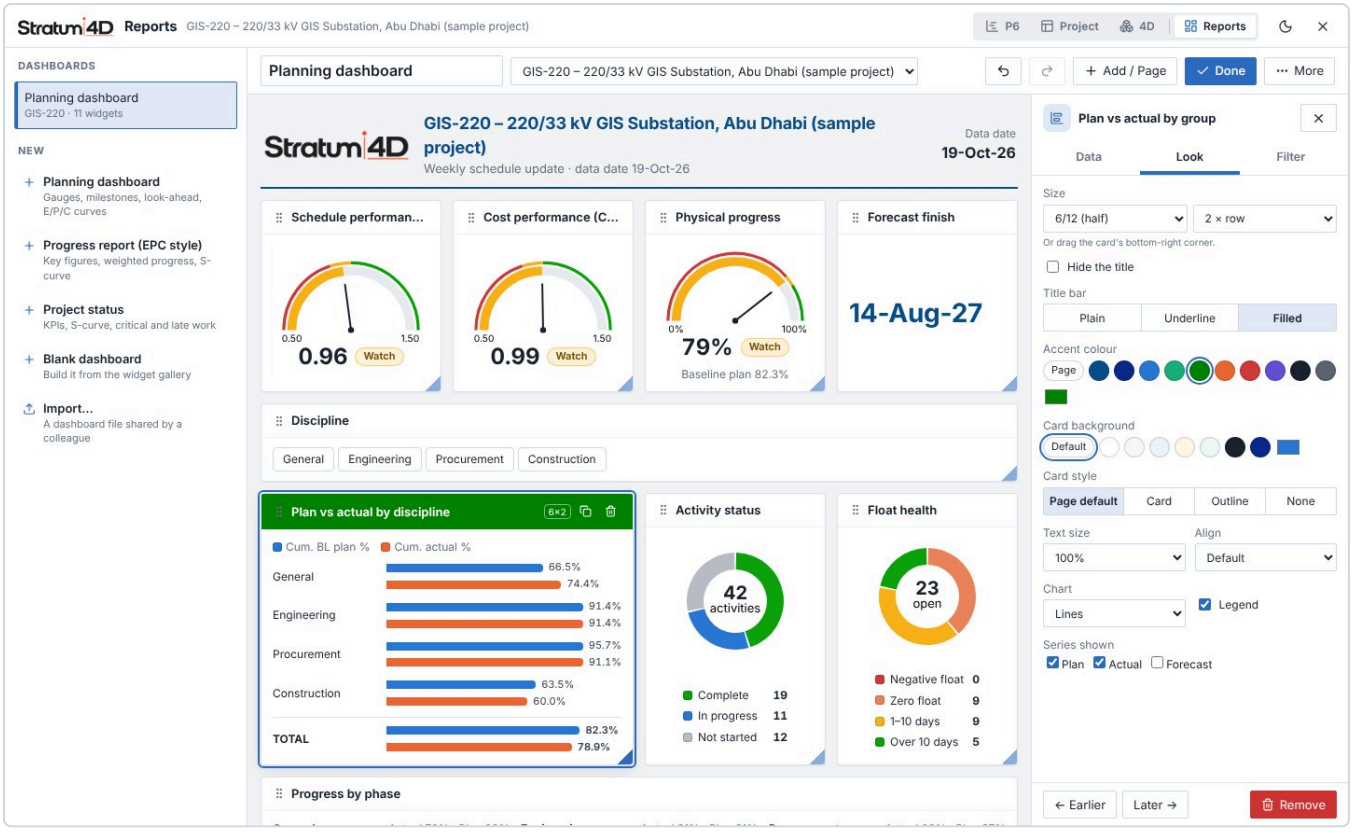


Figure 38. The Look tab for the plan-against-actual widget: a filled title bar in green, with the forecast series switched off.

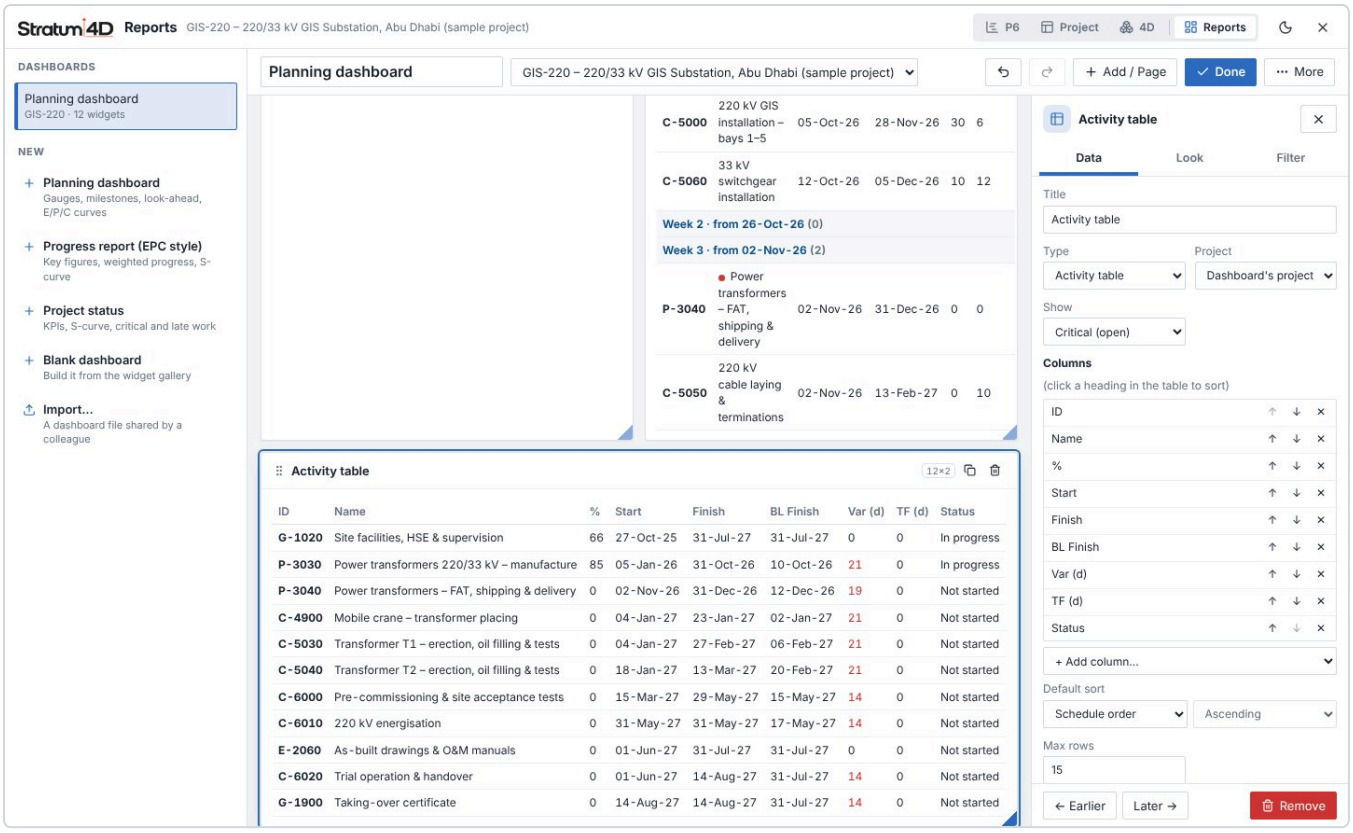


Figure 39. An activity table's Data tab: the column list with move and remove buttons, + Add column, default sort and maximum rows.

Page, theme and branding

With nothing selected, *Page & theme* sets the whole dashboard: theme (Stratum, EPC report or Executive dark), main colour, page background, card style, spacing, a header and subtitle with `{project}` and `{datadate}`, the data date at top right, a company logo, a footer, and A4 landscape or portrait for the PDF. Emailed PDFs use the same settings.

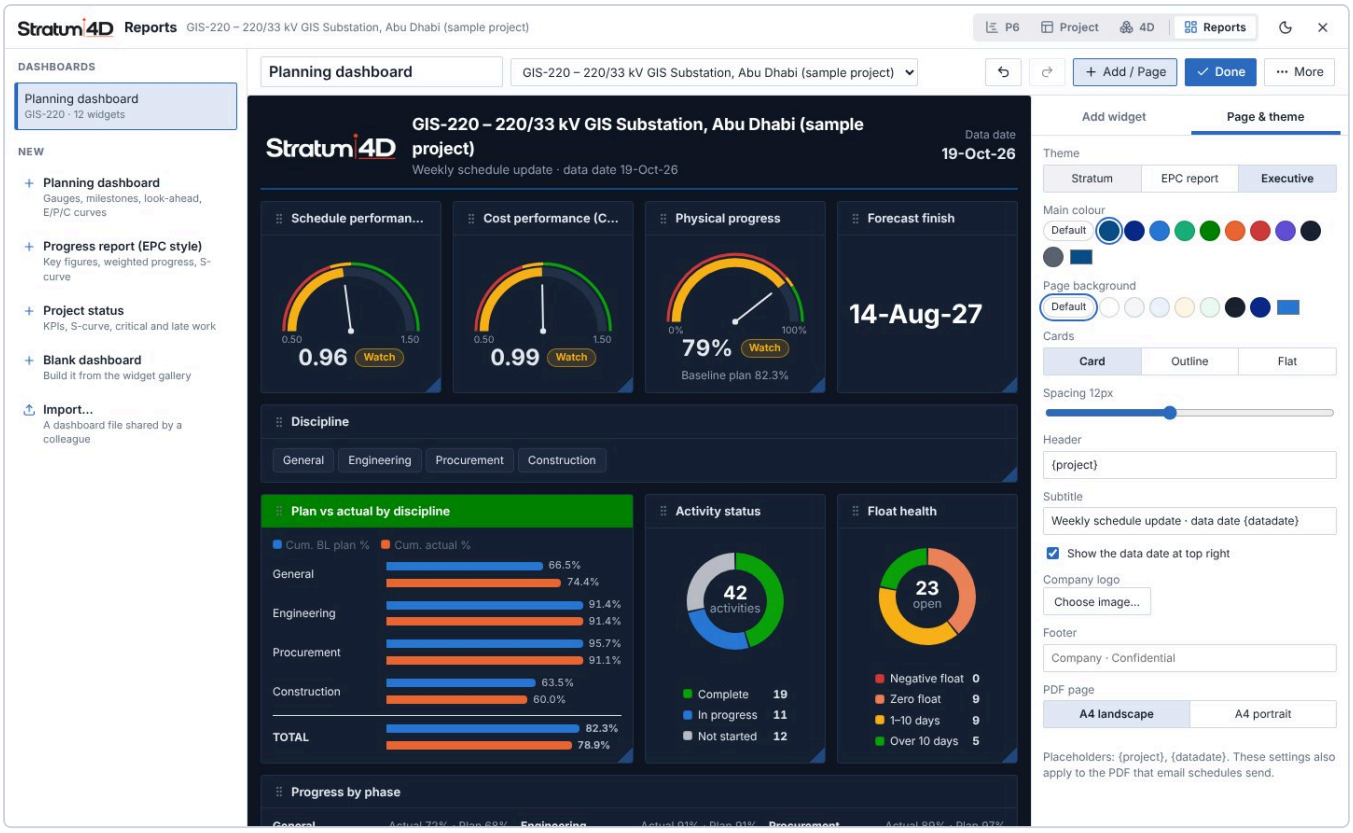


Figure 40. The same dashboard in the Executive theme, with the page settings on the right.

Manual weightage

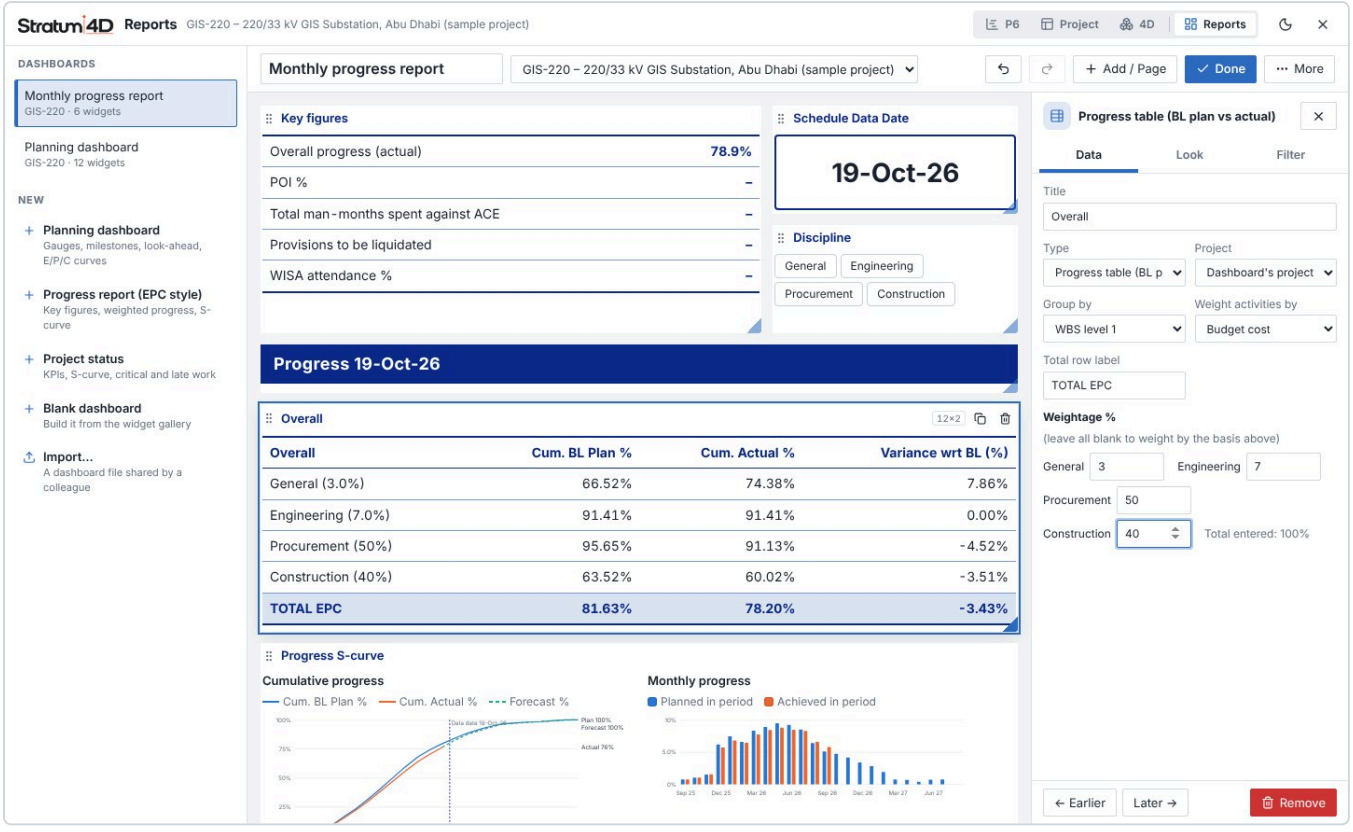


Figure 41. Selecting the progress table shows its weightage in the Data tab; here it is typed in as 3 / 7 / 50 / 40.

Typing a weightage per group overrides the calculated weights, so a report can follow the contract's agreed split. With 3 / 7 / 50 / 40 the example reports a cumulative plan of 81.63%, actual 78.20% and

variance -3.43%.

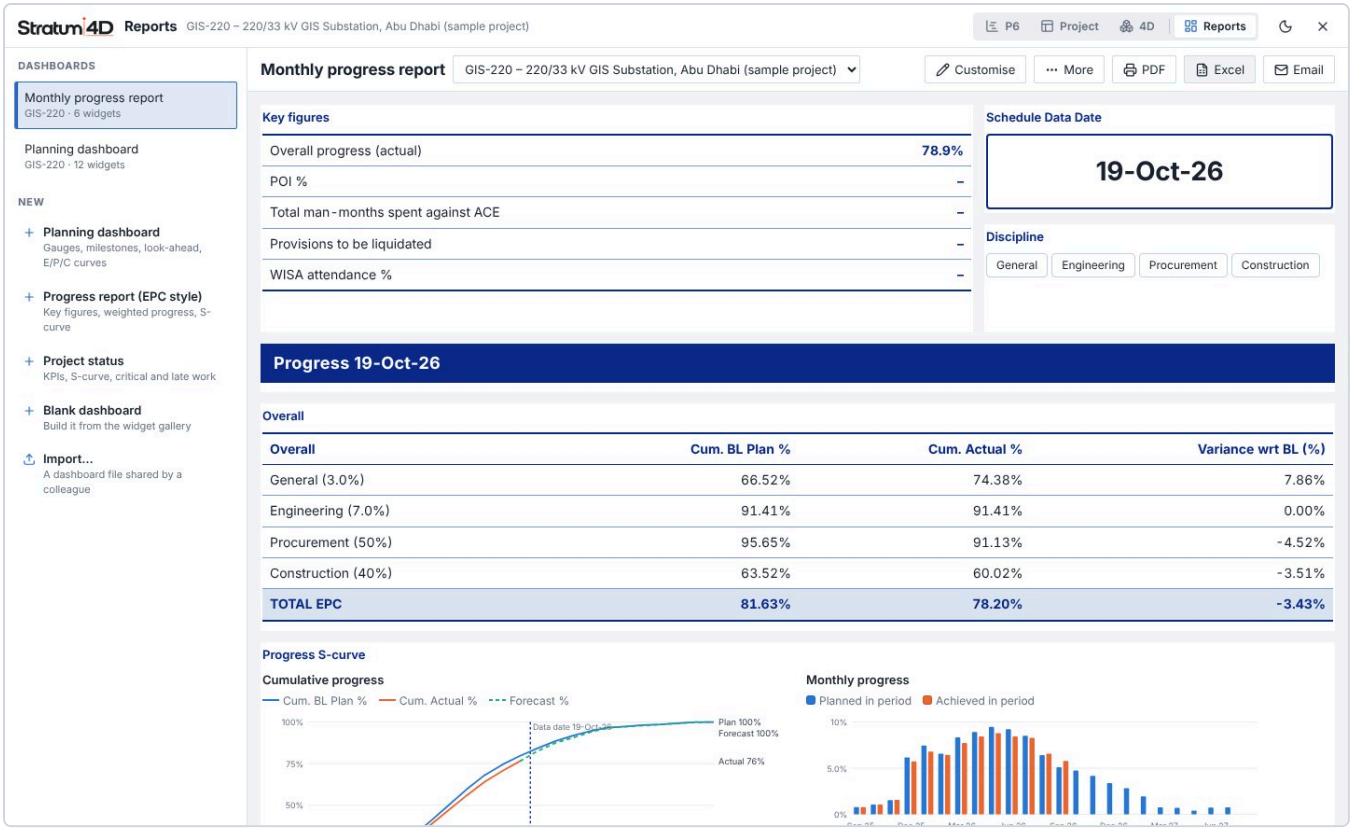


Figure 42. The report with the manual weightage: each row label shows its weight, and TOTAL EPC uses the typed split.

The key-figures table mixes calculated values (such as overall progress) with rows you type yourself, for figures that do not come from the schedule, for example POI %, man-months against ACE, provisions or WISA attendance.

Templates and sharing

More holds *Duplicate dashboard*, *Save as template* (it appears under *My templates* for everyone on the server), *Export dashboard file* (a .json file a colleague can bring in with *Import...*) and *Delete*.

Project status dashboard

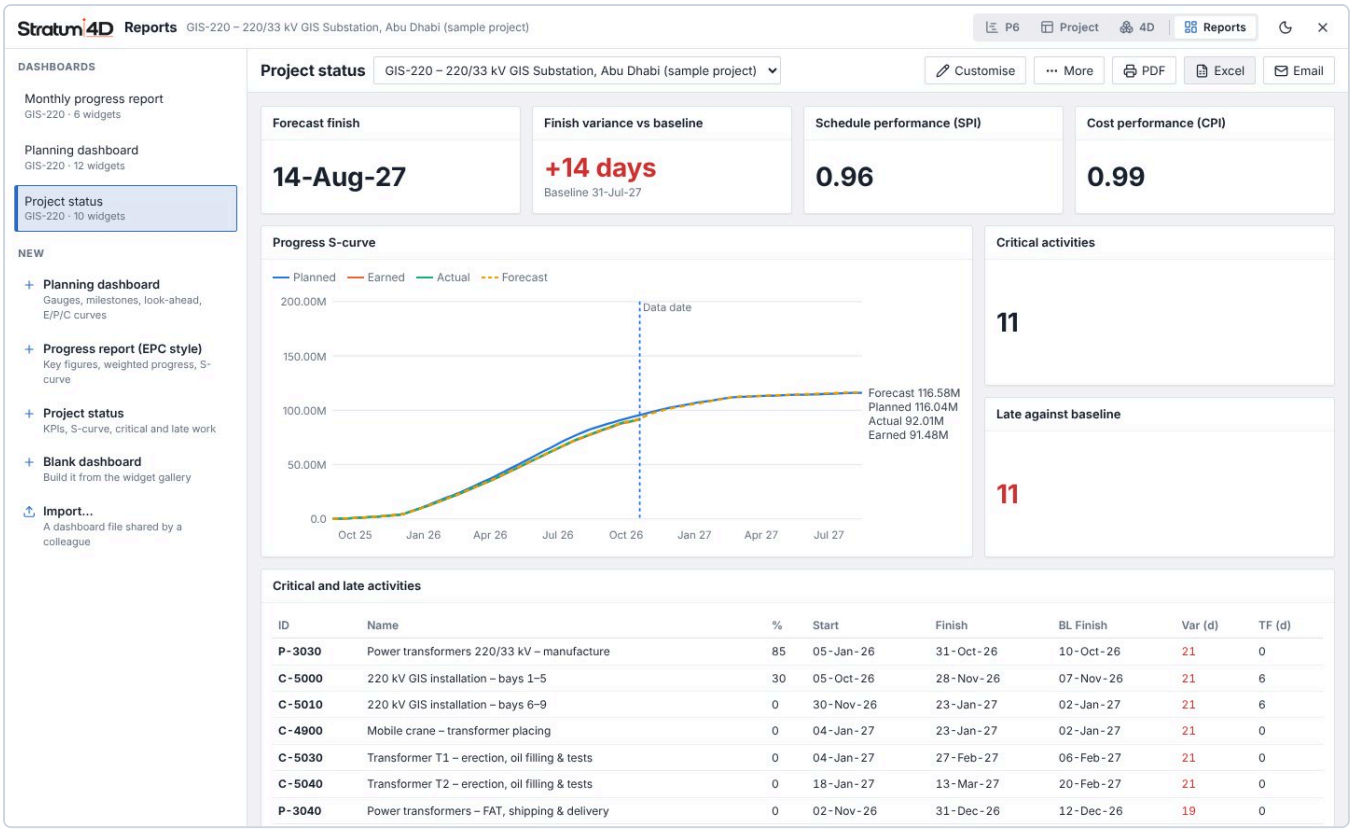


Figure 43. Project status template: forecast finish 14-Aug-27 (+14 days against the 31-Jul-27 baseline), SPI 0.96, CPI 0.99, the cost S-curve, critical and late counts, and the critical and late activities.

Group	Widgets
Numbers	KPI tile, gauge, key-figures table, status donut (progress status, float bands, critical share)
Charts	Cost or hours S-curve, progress S-curve with period bars, S-curve per phase, plan against actual by group, resource histogram, quantity curve
Tables	Activity table, milestone tracker, look-ahead (1–12 weeks), weighted progress table, portfolio of all projects
Layout	Section banner, text note, image or logo, slicer buttons

PDF and Excel

PDF prints the dashboard on A4 in the orientation set under *Page & theme*, with the page header and footer. *Excel* builds a workbook: summary, progress table, progress S-curve, late and critical, three-week look-ahead, all activities, S-curve data, weekly resources and portfolio.

Emailed reports

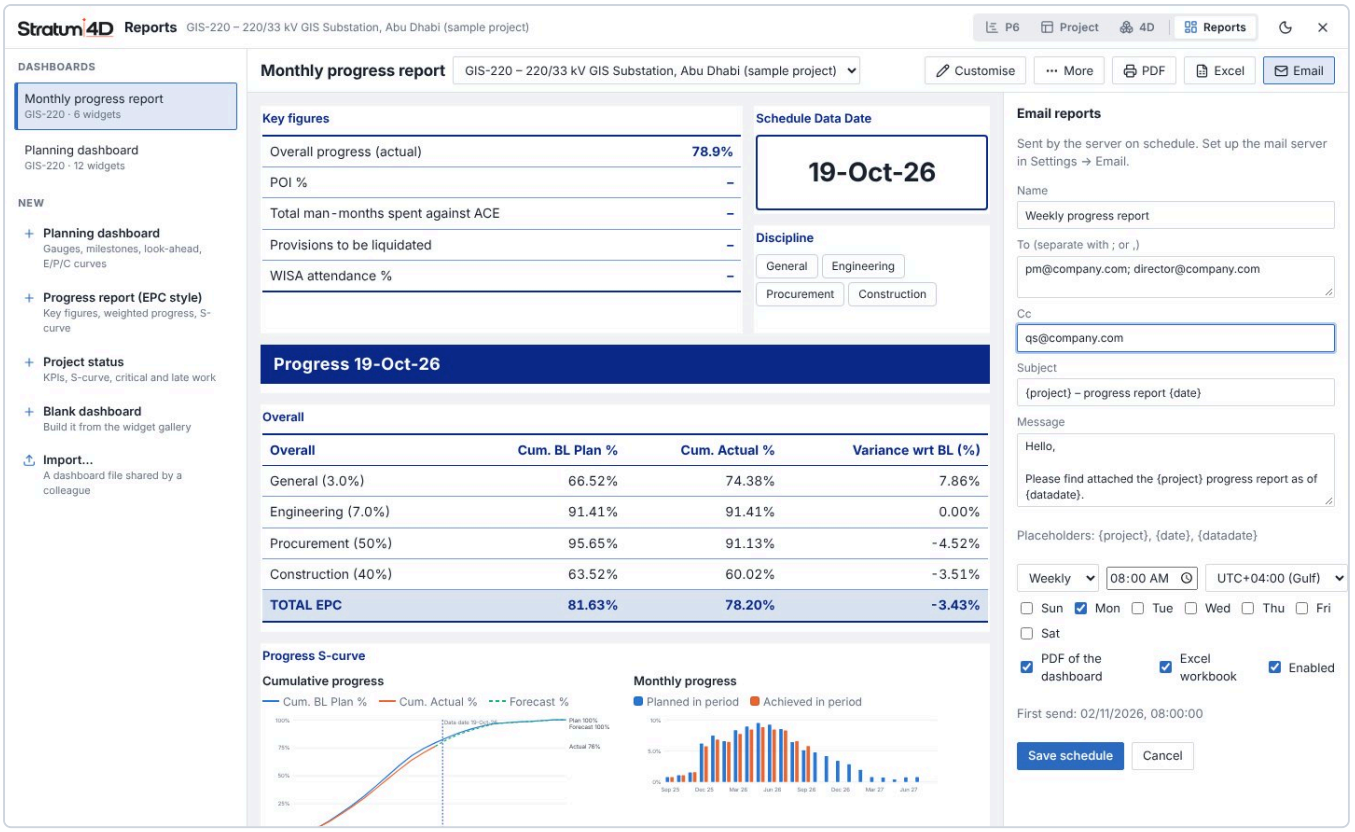


Figure 44. Email panel: a weekly schedule sending the report as PDF and Excel every Monday at 08:00 Gulf time to the project manager and director, copied to the QS.

Schedules run daily, weekly (on chosen days) or monthly (on a day of the month, or the last day). Subjects and messages accept `{project}`, `{date}` and `{datadate}`. Each run is logged with its result, and *Send now* sends immediately. On Stratum Server schedules run around the clock; in the desktop app they run while the app is open.

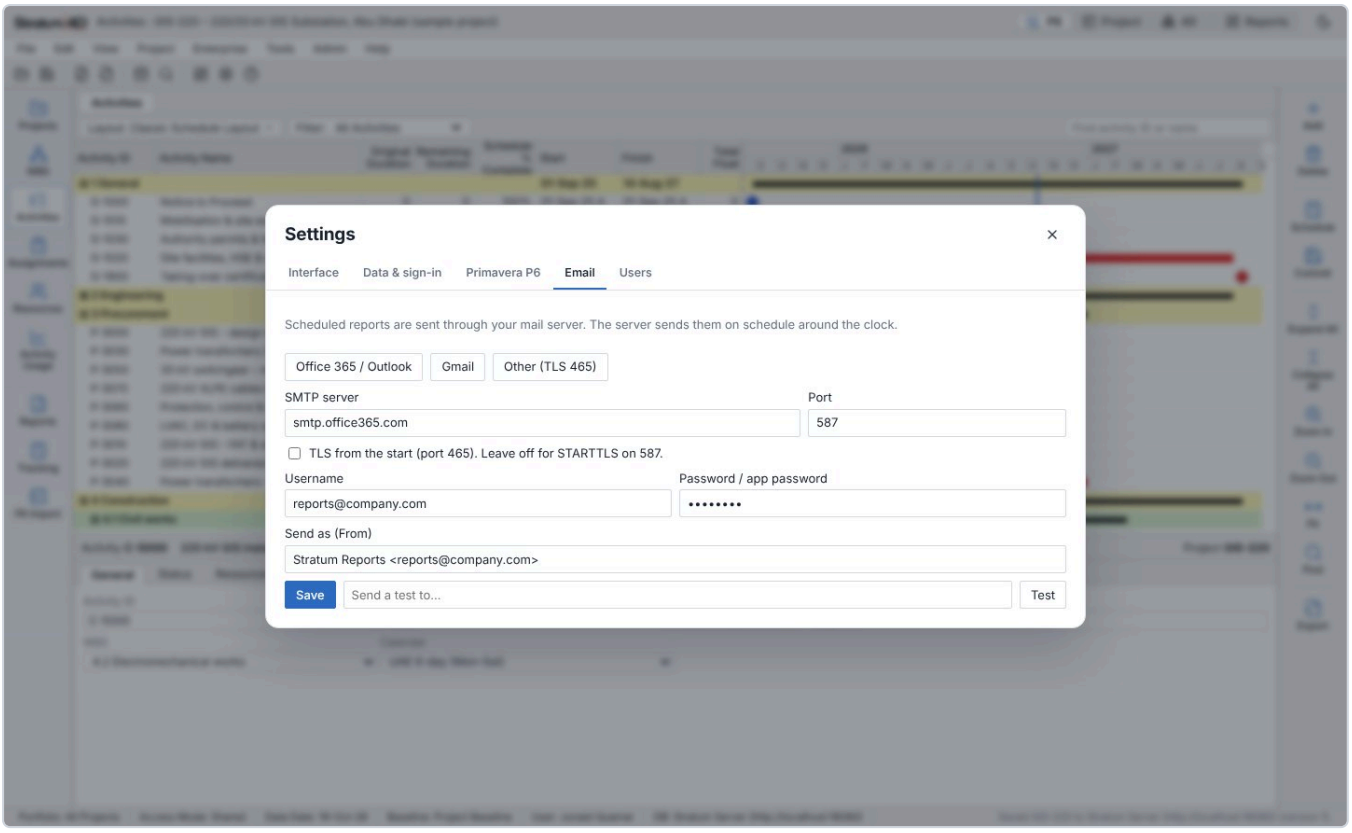


Figure 45. Settings → Email: the SMTP server (presets for Office 365 and Gmail), sender address and a test message.

11. Stratum Server for teams

Stratum Server is the shared version: one PostgreSQL database, sign-in, roles, P6 connections, scheduled emails and server-side PDF rendering. It ships as a Docker image with a Compose file that adds PostgreSQL 16.

```
cp .env.example .env      # set STRATUM_SECRET and POSTGRES_PASSWORD
docker compose up -d      # PostgreSQL 16 + Stratum Server on port 8080
```

Open `http://<server>:8080` and create the first administrator. In production, put the server behind your usual HTTPS front end (IIS, nginx or a load balancer).

Setting	Purpose
`STRATUM_SECRET`	Signs sign-in sessions and encrypts stored SMTP and P6 passwords. Keep it secret and do not change it later.
`DATABASE_URL`	PostgreSQL connection. Without it the server uses a SQLite file (`STRATUM_DB_FILE`).
`PORT` , `PUBLIC_URL`	Listening port (default 8080) and the address users see.
`NODE_EXTRA_CA_CERTS`	Company root certificate for reaching P6 over HTTPS.
`HTTPS_PROXY` , `NO_PROXY`	Outbound proxy for reaching P6.

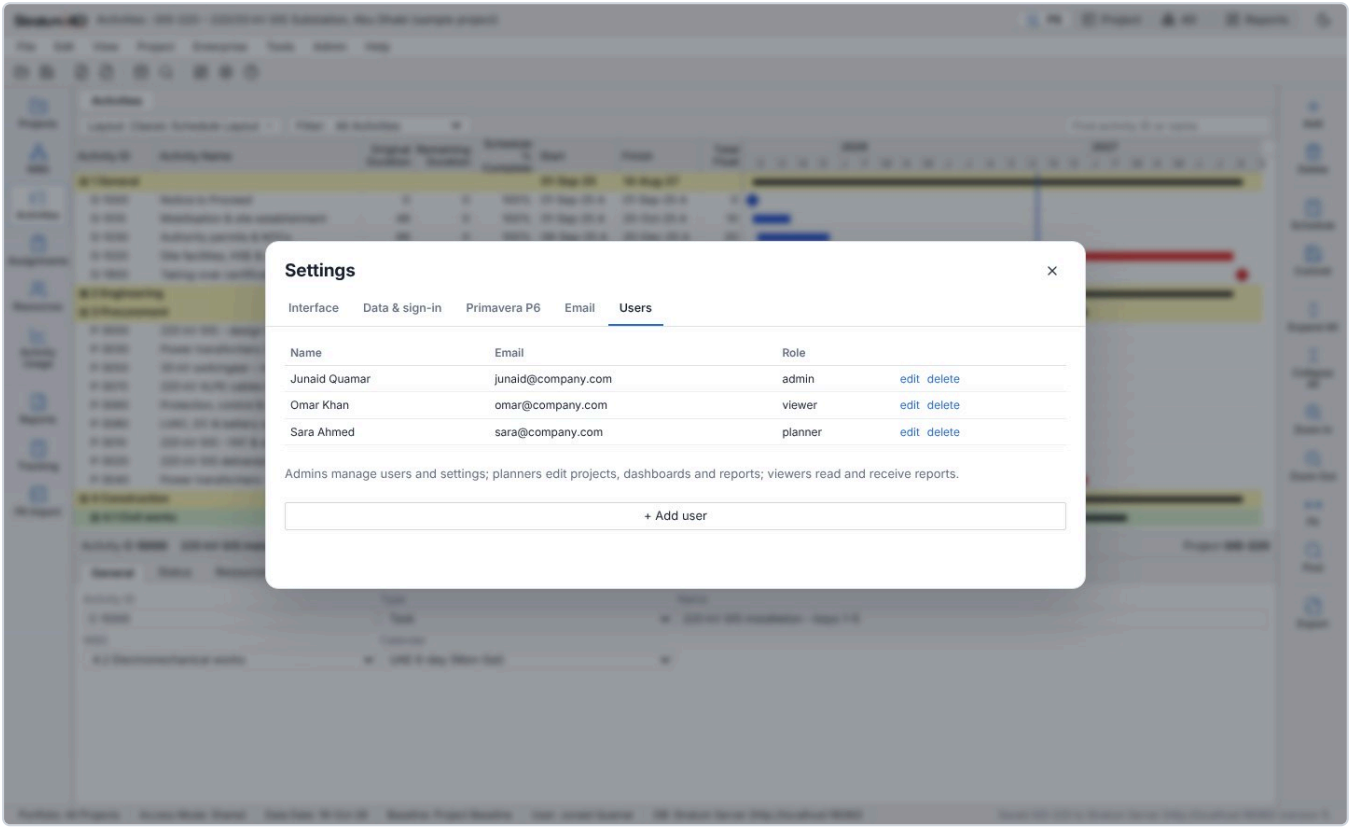


Figure 46. Settings → Users (administrators only): add users and give them a role.

Role	Can
Admin	Everything, including users, SMTP and P6 connection settings

Role	Can
Planner	Create and edit projects, dashboards and email schedules; import from P6
Viewer	Open projects and dashboards, and receive reports

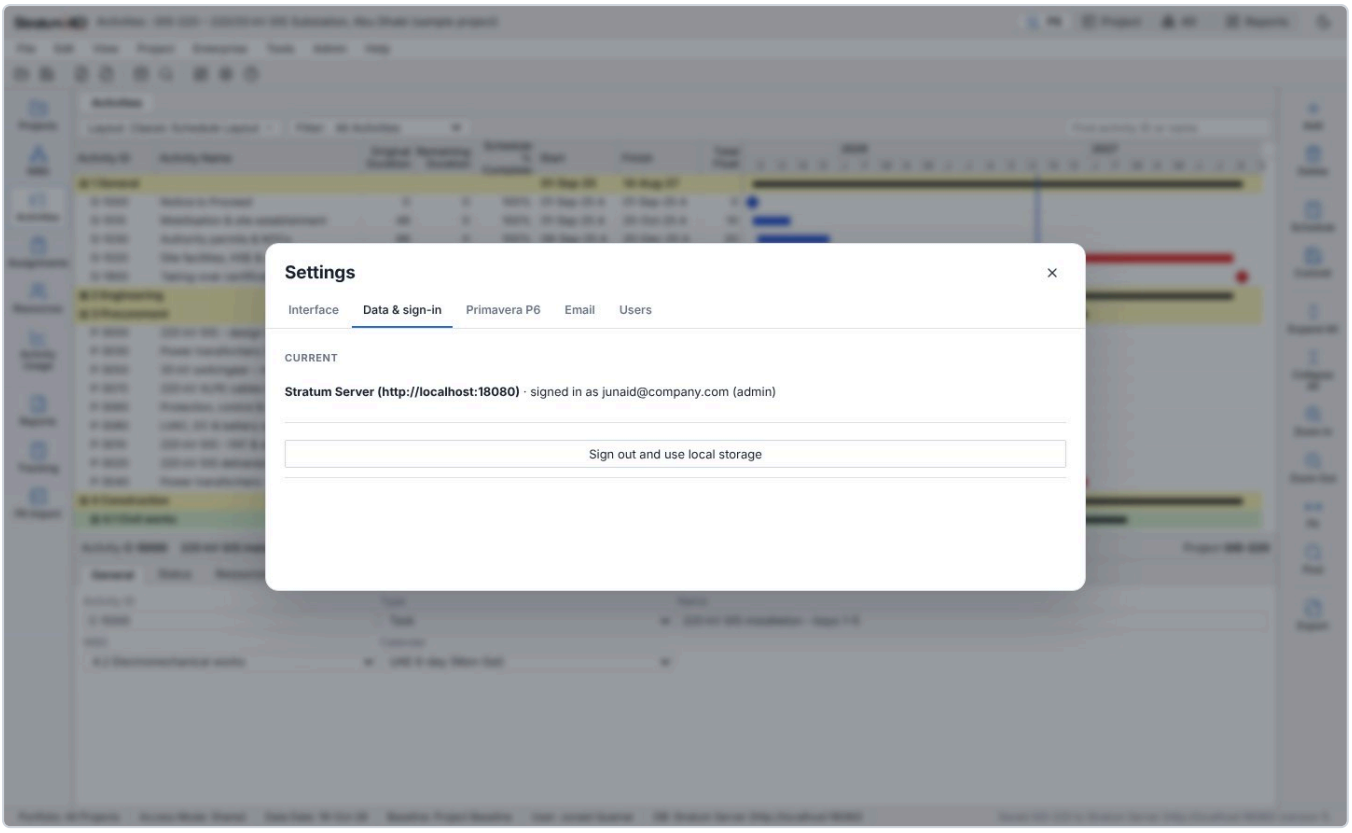


Figure 47. Settings → Data & sign-in: which database this client uses, and switching between local storage and a server.

12. Keyboard shortcuts

Keys	Action
`Space`	Play / pause
`←` `→`	Step one day
`Shift` + `←` `→`	Step one week
`Home` / `End`	Start / finish of the schedule
`F9`	Schedule (CPM)
`Ctrl` + `S`	Save (commit) to the database
`Ins` / `Del`	Add an activity after the selected one / delete the selected activity (P6 and MS Project layouts)
`Ctrl` + `F`	Find activities (P6 and MS Project layouts)
`Ctrl` + `Alt` + `1` / `2` / `3`	P6 / MS Project / 4D layout
`Ctrl` + `Alt` + `4`	Reports (dashboards)
`Ctrl` + `Z` / `Ctrl` + `Y`	Undo / redo while customising a dashboard
`Del` (dashboard)	Remove the selected widget while customising
Click, `Shift`-click in 3D	Select, add to selection
`Alt`-click an activity	Select its linked elements
Double-click an activity	Move the timeline to its start
`Esc`	Cancel path drawing, close dialogs
`?`	Getting-started help

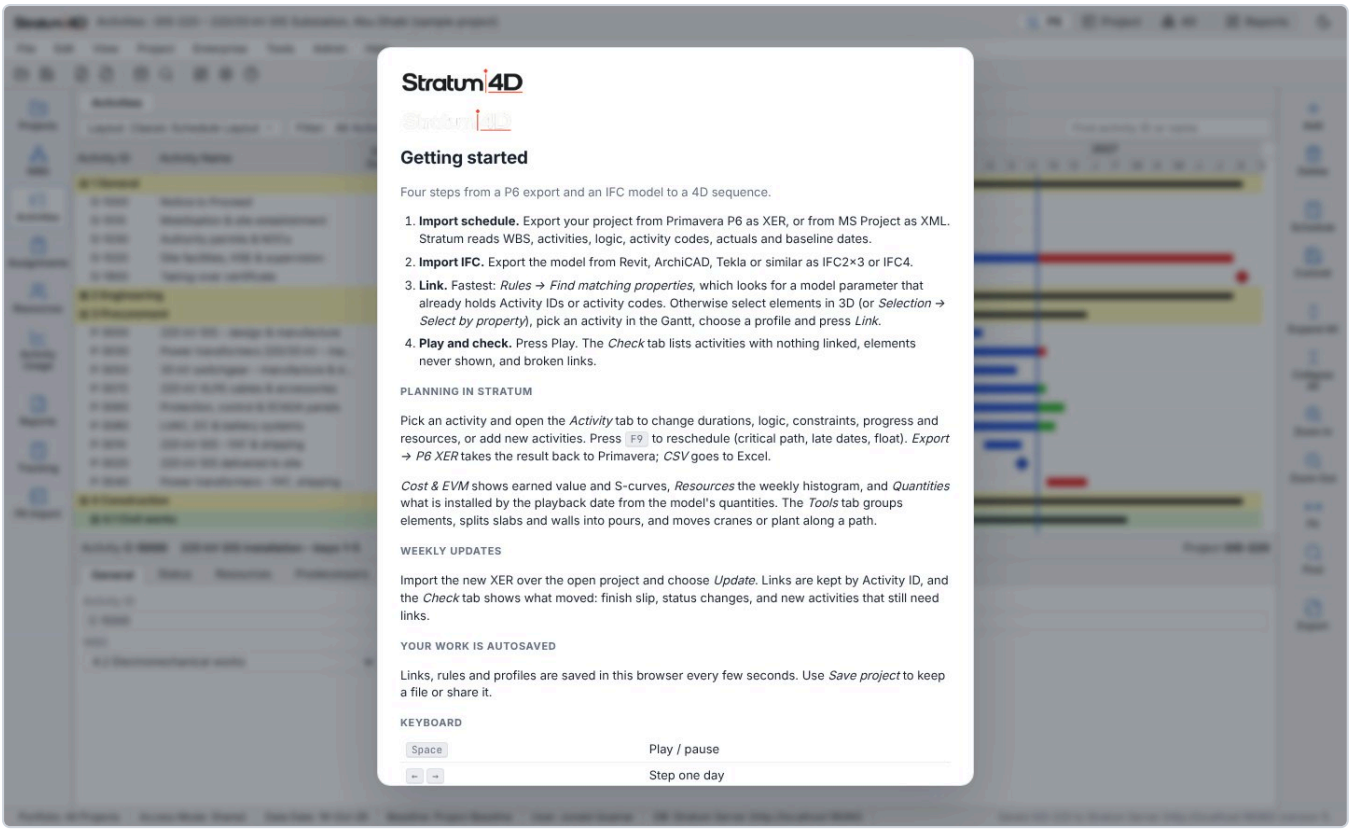


Figure 48. The getting-started help (?): four steps from a P6 export and an IFC model to a 4D sequence, plus shortcuts.

13. Troubleshooting

Symptom	Likely cause	What to do
P6 test stops at *Server name*	Not on the company network, or a typo	Connect to the VPN; check the address
P6 test stops at *Port open* with no answer	Firewall between this machine and P6	Ask IT to allow this computer (or the Stratum Server) to reach the P6 port
P6 test stops at *Certificate*	Company certificate authority not trusted	Install the root certificate in Windows/macOS, paste it into the connection, or set NODE_EXTRA_CA_CERTS on the server
P6 test stops at *P6 REST API* (404)	REST API not deployed at that path	Check the address ends in /p6ws/restapi; ask the P6 admin
Sign-in 401	Wrong user, password or database name	Check in the P6 web client; use the database name from the P6 login page
Sign-in 403	User has no API access	P6 Admin → Users → module access for web services / REST
No projects visible	No OBS access	Give the user project access in P6
“Not saved: someone else saved this project”	Another user saved a newer version	Use Save as new, or reopen the project and redo the change
Links lost after opening a project	The model is not stored in the database	Open the IFC again; links match by GlobalId
Emails not sent	SMTP settings, or the desktop app was closed	Send a test from Settings → Email; use Stratum Server for round-the-clock schedules
Windows protected your PC	Installer not code-signed yet	More info → Run anyway

14. Known limits

- The CPM engine follows P6's default options. Progress override, resource levelling and multiple float paths are not implemented yet.
- Costs and quantities are spread linearly over each activity, so weekly figures can differ slightly from P6's calendar-weighted spread.
- XER export has been checked by re-importing into Stratum, not yet against a live P6 database. Use P6's import preview the first time.
- The P6 EPPM connector has been tested against a mock of the documented REST API; the P6 database reader against SQL Server 2022 and SQLite copies of the P6 schema. Oracle has not been run against a live instance yet.
- Very large IFC files load slowly; GPU instancing and level of detail are planned.
- Undo/redo, a column chooser, group and sort, and in-app baseline management are on the roadmap. The P6 and MS Project layouts reproduce the arrangement of the originals; their menus contain only commands Stratum supports.
- Video export is WebM only.

Appendix A · Example project activities

All 46 activities of the example as imported (data date 19-Oct-26). OD = original duration in working days; TF = total float in days; ● = critical after F9. Baseline finish is shown only where it differs.

ID	Activity	WBS	OD	Start	Finish	BL finish	%	TF	Crit.
G-1000	Notice to Proceed	1	0	01-Sep-25	01-Sep-25		100	0	
G-1010	Mobilisation & site establishment	1	48	01-Sep-25	25-Oct-25		100	10	
G-1020	Site facilities, HSE & supervision	1	531	27-Oct-25	31-Jul-27		66	0	●
G-1030	Authority permits & NOCs	1	88	08-Sep-25	20-Dec-25		100	20	
G-1900	Taking-over certificate	1	1	14-Aug-27	14-Aug-27	31-Jul-27	0	0	●
E-2000	Basic design & single line diagram	2	60	08-Sep-25	15-Nov-25		100	0	
E-2010	Geotechnical & topographic survey	2	36	08-Sep-25	18-Oct-25		100	15	
E-2020	Civil & structural design – GIS building	2	87	20-Oct-25	31-Jan-26		100	0	
E-2030	Primary design – layouts, earthing, cable sizing	2	99	17-Nov-25	14-Mar-26		100	6	
E-2040	Secondary design – protection, control & SCADA	2	212	15-Dec-25	29-Aug-26	25-Jul-26	100	12	
E-2050	Client approval of detailed design	2	37	16-Mar-26	30-Apr-26	15-Apr-26	100	0	
E-2060	As-built drawings & O&M manuals	2	53	01-Jun-27	31-Jul-27		0	0	
P-3000	220 kV GIS – design & manufacture	3	198	01-Dec-25	31-Jul-26	11-Jul-26	100	0	
P-3010	220 kV GIS – FAT & shipping	3	47	03-Aug-26	26-Sep-26	05-Sep-26	100	0	
P-3020	220 kV GIS delivered to site	3	0	28-Sep-26	28-Sep-26	07-Sep-26	100	0	
P-3030	Power transformers 220/33 kV – manufacture	3	249	05-Jan-26	31-Oct-26	10-Oct-26	85	0	●
P-3040	Power transformers – FAT, shipping & delivery	3	50	02-Nov-26	31-Dec-26	12-Dec-26	0	0	●
P-3050	33 kV switchgear – manufacture & delivery	3	189	02-Feb-26	19-Sep-26		100	18	
P-3060	Protection, control & SCADA panels	3	198	06-Apr-26	28-Nov-26		70	8	
P-3070	220 kV XLPE cables & accessories	3	201	02-Mar-26	31-Oct-26		90	6	
P-3080	LVAC, DC & battery systems	3	162	04-May-26	14-Nov-26		60	14	
C-4000	Site clearance, levelling & temporary fencing	4.4.1	34	27-Oct-25	06-Dec-25		100	0	
C-4010	Piling – GIS building	4.4.1	53	08-Dec-25	07-Feb-26	31-Jan-26	100	0	
C-4020	Raft foundation – GIS building	4.4.1	39	09-Feb-26	28-Mar-26		100	0	
C-4030	GIS building RC frame	4.4.1	67	30-Mar-26	20-Jun-26		100	0	
C-4040	GIS building roof steel & cladding	4.4.1	59	22-Jun-26	29-Aug-26	15-Aug-26	100	0	
C-4050	EOT crane installation & load test	4.4.1	18	31-Aug-26	19-Sep-26		100	0	
C-4060	GIS building finishes & HVAC	4.4.1	72	07-Sep-26	28-Nov-26		45	10	
C-4070	Control & 33 kV building – foundations & structure	4.4.1	118	16-Feb-26	11-Jul-26		100	20	
C-4080	Transformer foundations, oil pits & firewall	4.4.1	97	06-Apr-26	01-Aug-26		100	30	
C-4090	Cable trenches & ducts	4.4.1	150	04-May-26	31-Oct-26		75	4	
C-4100	Roads, drainage, boundary wall & gatehouse	4.4.1	141	14-Sep-26	27-Feb-27		15	40	
C-4900	Mobile crane – transformer placing	4.4.1	18	04-Jan-27	23-Jan-27	02-Jan-27	0	0	●
C-5000	220 kV GIS installation – bays 1–5	4.4.2	48	05-Oct-26	28-Nov-26	07-Nov-26	30	6	
C-5010	220 kV GIS installation – bays 6–9	4.4.2	45	30-Nov-26	23-Jan-27	02-Jan-27	0	6	

ID	Activity	WBS	OD	Start	Finish	BL finish	%	TF	Crit.
C-5020	220 kV GIS HV testing	4.4.2	24	25-Jan-27	20-Feb-27		0	6	
C-5030	Transformer T1 – erection, oil filling & tests	4.4.2	48	04-Jan-27	27-Feb-27	06-Feb-27	0	0	•
C-5040	Transformer T2 – erection, oil filling & tests	4.4.2	45	18-Jan-27	13-Mar-27	20-Feb-27	0	0	•
C-5050	220 kV cable laying & terminations	4.4.2	87	02-Nov-26	13-Feb-27		0	10	
C-5060	33 kV switchgear installation	4.4.2	46	12-Oct-26	05-Dec-26		10	12	
C-5070	Protection, control & SCADA panels installation	4.4.2	77	07-Dec-26	06-Mar-27		0	4	
C-5080	Earthing & lightning protection	4.4.2	214	06-Apr-26	19-Dec-26		70	30	
C-5090	LVAC, DC & battery installation	4.4.2	47	07-Dec-26	30-Jan-27		0	14	
C-6000	Pre-commissioning & site acceptance tests	4.4.3	63	15-Mar-27	29-May-27	15-May-27	0	0	•
C-6010	220 kV energisation	4.4.3	0	31-May-27	31-May-27	17-May-27	0	0	•
C-6020	Trial operation & handover	4.4.3	65	01-Jun-27	14-Aug-27	31-Jul-27	0	0	•

Appendix B · Terms in each layout

Concept	P6 style	MS Project style	4D style
Activity	Activity	Task	Task
Activities	Activities	Tasks	Tasks
Activity Id	Activity ID	ID	Task ID
Activity Name	Activity Name	Task Name	Task Name
Wbs	WBS	Outline	WBS
Data Date	Data Date	Status Date	Data Date
Total Float	Total Float	Total Slack	Total Float
Free Float	Free Float	Free Slack	Free Float
Original Duration	Original Duration	Duration	Planned Duration
Remaining Duration	Remaining Duration	Remaining Duration	Remaining Duration
Percent Complete	Physical % Complete	% Complete	% Complete
Predecessors	Predecessors	Predecessors	Predecessors
Successors	Successors	Successors	Successors
Relationship	Relationship	Link	Dependency
Baseline	Baseline	Baseline	Baseline
Schedule	Schedule	Calculate	Reschedule
Resources	Resources	Resource Names	Resources
Constraint	Primary Constraint	Constraint Type	Constraint
Milestone	Milestone	Milestone	Milestone
Project	Project	Project	Project
Projects	Projects	Projects	Projects
Playback Date	Timescale date	Date	Focus date

Stratum 4D

Stratum 4D · User Guide · Release 0.3 · October 2026

Conceived, designed and developed by **Junaid Quamar**, Senior Planning Engineer.
User guide written and prepared by Junaid Quamar.
linkedin.com/in/junaidquamar

The worked example (220/33 kV GIS Substation, Abu Dhabi) is fictional; names, quantities and values are illustrative. Fonts: Inter, IBM Plex Sans, Source Sans 3, Nunito Sans and JetBrains Mono under the SIL Open Font License.